COMPLETE THESIS WRITING GUIDE

DynaCampus-FR

A self-contained teammate handbook aligned to the original
United International University, Department of CSE FYDP template.

Proposed thesis title

Few-Photo Enrollment and Open-Set Face Recognition under Masking and Controlled Degradation: An Experimental Study

Prepared 23 October 2026. This PDF supplies the writing plan, adaptable prose, methods, diagrams, numerical tables, software checks, discussion, engineering mappings and references in one place. It is a guide for preparing the thesis, not an institutionally approved thesis or a report of new experiments.

What is inside	How to use it
Template-complete guidance	Follow the original template headings reproduced in the contents. The earlier report's changed numbering is not used.
Copyable diagrams and tables	Figures and tables are embedded. Copy or redraw the embedded figures; captions and placement guidance are supplied.
Verified research evidence	Main final, development and supplementary results are kept distinct. No new final image inference was performed.
Explicit author inputs	Only personal/institutional facts, real spending and actual contribution records require author confirmation. Nothing is fabricated to fill them.

Research snapshot: 55 final gallery identities; primary correct known acceptance 151/160 (94.38%); 0/151 observed unknown false accepts. Full context, uncertainty and limitations appear in Chapter 4.

Reader map and template coverage

Use this handbook without opening repository files. Evidence identifiers are included for traceability; all writing-critical facts and selected complete result tables are reproduced here. Reference URLs identify sources for formal citation, not missing instructions.

The supplied original template has 17 physical pages. It requires a single-paragraph abstract; acknowledgements; an optional publication list; contents, figure and table lists; six chapters; and references. Each chapter starts with one or two outline sentences. Chapter 3 requires alternative solutions and selection rationale. Chapter 5 requires alternatives/pros/cons for relevant standards, an alternative budget and revenue model, P1-P7 mappings with a knowledge-profile mapping for P1, and A1-A5 mappings with explanatory subsections. This guide covers those items explicitly. [T1]

The template's political constraint, communication standards, preliminaries, similar applications, UI design and chapter summaries must not disappear simply because the previous report omitted or renamed them. The publication list is explicitly optional. Approval, declaration, plagiarism forms and a fixed word/page limit are not specified in the supplied PDF; add them only if your department separately requires them.

Scope label	Meaning	Where used
FINAL	Methods and thresholds frozen before the reserved original-image evaluation; local identity-disjoint roles.	Main halfFace comparison
DEVELOPMENT	Calibration and assessment reuse examined development images; descriptive findings.	Backbones, decisions, pooling, stress, enrollment variants and gallery growth
EXPLORATORY	Custom sequence-separated crop study; prior pilot exposure and repeated identities.	ChokePoint supplement
ENGINEERING	Executable functionality, persistence, latency or software checks; not a population accuracy estimate.	ICFD sample, dashboard, unit tests and runtime

Writing sequence. Draft Chapter 3 and Chapter 4 from this guide, then Chapter 2 and Chapter 5. Align the objectives and conclusion next, and write the final abstract last. Preserve heading numbering. Let your thesis editor generate page numbers, contents and figure/table lists automatically.

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Front matter: title, abstract and preliminary pages

Title page

Retain the official title-page arrangement, UIU logo, degree wording, department and institution. The new title is a proposal for supervisor approval, chosen to describe the evaluated scope rather than imply a validated campus rollout.

- Use the proposed title on the cover and consistently in the abstract and conclusion. Keep DynaCampus-FR as the software/system name.
- Retain: "Submitted in partial fulfilment of the requirements of the degree of Bachelor of Science in Computer Science and Engineering", Department of Computer Science and Engineering, United International University.
- Use the actual final submission date; the template's 22 October 2026 is an example, not evidence of submission.
- Do not claim lightweight deployment from model naming or real-time throughput from an unbenchmarked camera page.

Section completion check: Title, author names/IDs, submission date and institutional layout agree everywhere.

Author shown in previous report	ID shown there - confirm with student record
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These names and IDs are transcribed from the previous report, not independently verified. Do not infer a supervisor name, signature or approval. Actual member contribution records must be supplied in Section 3.4.

Abstract

The original template explicitly requires ONE PARAGRAPH covering background, methodology and results. The following draft meets that structural requirement. It can be adapted after supervisor review; retain its scope boundaries.

Ready-to-adapt text.

> Face recognition for institutional monitoring must identify enrolled individuals while refusing unknown or ambiguous matches when enrollment photographs are limited and probe images are degraded. This study investigates enrollment and rejection choices around frozen pretrained face-recognition models using a SCRFD detection and alignment pipeline, normalized embedding templates and calibrated open-set decisions. An audited halfFace dataset was partitioned into development and reserved identity groups, and the final methods were fixed before reserved-image evaluation. One-photo score-only enrollment correctly accepted 130 of 160 masked known probes, compared with 150 of 160 for up-to-three-photo enrollment. The primary few-photo score-and-margin method accepted 151 of 160 known probes (94.38%) with no observed false acceptance among 151 unknown probes; its one-probe increment over few-photo score-only decisions had an uncertainty interval including zero. Additional ArcFace and AdaFace development experiments found no consistent benefit from the tested synthetic enrollment profile or quality-weighted pooling. In an exploratory ChokePoint supplement, five-frame averaging improved severe-degradation Rank-1 retrieval from 295/303 to 302/303, while mixed-track thresholds falsely accepted 12/143 unknown queries, demonstrating incomplete calibration transfer. A persistent gallery prototype and CPU latency measurements complement the experiments. The contribution is reproducible evidence about existing recognition components and their operating choices; representative campus deployment, live tracking performance and population-level zero-error claims remain unestablished.

Section completion check: One paragraph; no FYDP1 proposal-only wording; denominators and evidence scopes retained.

Suggested keywords: open-set face identification; few-photo enrollment; ArcFace; AdaFace; masked face recognition; temporal aggregation; calibration. Keywords are an editorial addition, not an explicit template requirement.

Acknowledgements

Write in a consistent first-person plural voice. Thank only real people or organizations. Fill the supervisor names and specific support from actual records. Do not imply institutional approval of surveillance or data collection.

Ready-to-adapt text.

> We thank our supervisor, [confirmed supervisor name and designation], for guidance and feedback throughout the project. We acknowledge [confirmed laboratory, department or individuals] for [specific support actually received]. We also recognize the authors and maintainers of the public datasets and software used in this study. Finally, we thank our families and teammates for their support. The authors remain responsible for the analysis, interpretation and reporting of the results.

Section completion check: Replace author-input brackets with true facts, or remove the inapplicable sentence; do not fabricate names.

Publication List - optional

The template distinguishes journal articles, conference papers and additional relevant publications. No accepted/published work is established by the current project artifacts. If there are none, omit this optional page, or state "No publications to report" if your supervisor wants it retained. A thesis draft is not an accepted publication. Do not list a paper as "in preparation" unless the team is actually preparing that named manuscript.

Contents, List of Figures and List of Tables

Generate all three lists from real heading/caption fields. The supplied template uses Roman numerals for preliminary pages and Arabic numerals from Chapter 1. Follow its typography and institutional layout; the guide's own typography is a reading aid, not a replacement style specification.

- Remove placeholder instructions and example Algorithm Design reference from the final thesis unless genuinely cited.
- Use one caption per numbered figure/table; cite each in the narrative before or near its placement.
- Keep chapter title and numbering exactly as the original template: 2.1 Preliminaries, 2.2 Literature Review, 2.3 Gap Analysis; 4.2 Testing and Evaluation, 4.3 Results and Discussion.
- Guide diagrams are proposed figure slots. If figures are inserted or removed, regenerate thesis numbering automatically.

Chapter 1. Introduction

Ready-to-adapt text.

> This chapter introduces the institutional face-recognition problem, motivates the investigation of limited enrollment and unknown-person rejection, and states the objectives and evaluated scope. It also outlines the methodology, project outcomes and organization of the report.

1.1 Project Overview

Build three connected paragraphs: institutional motivation; technical recognition problem; concrete study scope. Distinguish identity classification from open-set identification.

- Motivation: distant, masked or degraded observations make matching uncertain, and an unenrolled visitor must not automatically inherit the nearest registered identity.

- Problem statement: with frozen pretrained recognizers and few enrollment views, determine how templates and rejection rules affect correct identity acceptance, unknown acceptance and refusals.

- Scope: original masked public images, descriptive development studies, a custom crop-based temporal supplement and a persistent Streamlit prototype. No collected local campus cohort or deployed access-control installation is claimed.

Ready-to-adapt text.

> DynaCampus-FR investigates how a small set of enrollment photographs can support open-set face identification. Unlike a fixed classifier whose output classes are its trained identities, the implemented system stores templates computed by a pretrained recognizer and searches an expandable gallery. The experimental problem is to retain correct enrolled-person matches while rejecting unknown or ambiguous probes, with explicit accounting for capture failures. Educational CCTV motivates the project, but the evidence reported here comes from public images, controlled transformations and a separate temporal crop study.

1.2 Motivation

Explain the computational reasons for the study and the potential benefit. Avoid unsupported demographic claims or claims that manual attendance has already been replaced.

- Few available photos make enrollment policy a practical design choice; additional original views and synthetic variants may behave differently.

- Retrieving a close template does not establish enrollment membership. Thresholds, ambiguity and coverage must be considered together.

- Quality cues can correlate with similarity without producing better aggregation; test rather than assume their usefulness.

- Image processing can dominate small-gallery search cost; measured timing helps identify the actual bottleneck.

- Potential benefit is evidence to inform future systems, not demonstrated reductions in crime, proxy attendance, bias or staff workload.

1.3 Objectives

Enumerate specific, testable objectives and map each to a reported outcome.

- **O1:** Audit original/derived images, resolve documented exclusions and establish frozen identity roles.

- **O2:** Compare one-photo and up-to-three-photo enrollment at development-selected open-set operating points on a reserved original-image evaluation.

- **O3:** Examine score, margin and quality refusal rules and report acceptance, false identification, coverage and failures.
- **O4:** Compare released ArcFace/AdaFace checkpoints, pooling and enrollment-variant policies on development data, and test fixed-probe gallery composition.
- **O5:** Investigate temporal averaging, quality proxies, simulated identity contamination and threshold transfer in the exploratory ChokePoint study.
- **O6:** Demonstrate persistent training-free enrollment and measure CPU image-processing, insertion, search and decision latency.

Section completion check: Do not retain full live deployment, edge-device benchmarking, fairness or anti-spoofing as achieved objectives.

1.4 Methodology

Summarize the sequence rather than duplicate Chapter 3 equations.

Ready-to-adapt text.

> The methodology first audits public source images and freezes identity roles, then processes the main image inputs through SCRFD detection, five-point alignment and a frozen ArcFace recognizer. One-photo and few-photo normalized gallery templates are compared using development-calibrated score and margin rules. After settings are frozen, reserved original images are evaluated once. Additional development studies examine checkpoint choice, pooling, synthetic enrollment, degradation and gallery composition. A separate ChokePoint experiment uses supplied aligned face crops to investigate five-frame aggregation and calibration transfer. Software tests, saved prediction audits, a gallery demonstration and CPU measurements support implementation verification.

1.5 Project Outcome

List actual deliverables and the nature of the contribution.

- A modular inference pipeline, persistent gallery and working still-image enrollment/recognition workflow.
- An identity-disjoint main comparison showing a large observed few-photo enrollment benefit and a smaller uncertain incremental margin effect.
- Negative/mixed findings for the tested quality/augmentation choices, plus calibration-transfer and contamination failure cases.
- Recorded configurations, source/checkpoint hashes, scores, paired uncertainty and separate recomputation scripts.
- Measured CPU latency within stated software boundaries. No production deployment or novel recognition architecture is established.

1.6 Organization of the Report

Use one narrative paragraph with all six chapters.

Ready-to-adapt text.

> Chapter 2 explains the preliminary concepts, similar applications, related research and research gap. Chapter 3 presents the requirements, system models, detailed methods, alternative designs, project plan and task allocation. Chapter 4 describes the environment, testing protocols, results and discussion. Chapter 5 examines relevant standards, design constraints, costs and complex engineering mappings. Chapter 6 summarizes the contributions, limitations and directions for future work.

Chapter 2. Background

Ready-to-adapt text.

This chapter introduces the concepts needed to interpret template-based open-set recognition, reviews related applications and research, and identifies the focused gap addressed by the experiments. The review distinguishes established methods from the project's experimental contribution.

2.1 Preliminaries

Supply the following concepts before the literature review. They are essential for readers unfamiliar with face recognition.

Term	Meaning in this thesis
Detection / alignment	Find face boxes and landmarks; geometrically standardize a face crop. Neither alone identifies a person.
Embedding / template	A numerical descriptor from a pretrained model; a stored identity representation may average several descriptors.
Verification / identification	Verification compares a claimed pair; identification searches a gallery. Their false-match denominators differ.
Closed set / open set	Closed set assumes a gallery mate; open set includes people absent from the gallery and requires rejection.
Enrollment / backbone training	Enrollment stores templates; training changes model weights. Training-free enrollment does not mean the pretrained model was never trained.
Score / margin / quality	Similarity to a template; best-minus-second similarity gap; an input quality proxy. A cosine score is not a calibrated probability.
Calibration / evaluation	Choose settings on development data, then assess frozen settings on separate inputs. Reused development assessment is descriptive.
Coverage / failure	Coverage counts decisive KNOWN or UNKNOWN outcomes; ambiguity and low-quality refusal are withheld decisions. Capture failure occurs before matching.
Synthetic degradation / independent data	A transformation tests sensitivity; it does not add new people, independent views or an independent test cohort.
Paired cluster interval	Resample identities with their observations, comparing methods on the same probes; uncertainty still conditions on chosen thresholds.

2.2 Literature Review

Separate software application comparisons from research papers. Summarize method, setting, relevance and limitation. Do not combine published scores from different datasets into an accuracy ranking of your system.

2.2.1 Similar Applications

Use actual application/tool examples and compare functions, not untested comparative accuracy. The primary project pages were checked for these features. [R9-R10]

Application/tool	Relevant functionality	Relationship to DynaCampus-FR
CompreFace	An open-source face-recognition service with API-based integration and subject/image management.	Shows that service-oriented enrollment/recognition already exists. This project studies enrollment and rejection experiments; no benchmark against CompreFace was run.
DeepFace	Python recognition/verification tooling with multiple model options and service/web integration examples.	Provides broader integration context. No claim that DynaCampus-FR is more accurate, faster or safer; do not conflate a framework with this project's frozen protocol.
DynaCampus-FR prototype	Streamlit enrollment, still-image recognition, gallery management and separate evidence viewer.	Functional prototype with recorded demonstrations; no institutional authentication, attendance backend or live-camera deployment benchmark.

2.2.2 Related Research

Use the following compact synthesis as the core literature table. Expand the prose by explaining the connection to your own experimental choices. [R1-R8]

Reference	Established idea / relevance	Boundary for this project
ArcFace [R1]	Angular-margin-trained face embeddings.	Use a released checkpoint; no new angular loss is trained.
AdaFace [R2]	Quality-related adaptive training margin using feature norm as a proxy.	Released IR-50 comparator; loss and pretraining effects are not isolated.
SCRFD [R3]	Efficient face-detection family.	One detector checkpoint used; no new WIDER FACE AP evaluation.
Quality pooling [R4]	Prior template-quality pooling work.	Mean/weighting are baselines, not inventions; empirical utility must be checked.
SORT [R5]	Kalman/association-based online tracking.	SORT-like prototype exists; supplementary tracks are supplied by labels, not measured tracker output.
halfFace [R6]	Masked/unmasked South Asian image dataset with original and derived collections.	Original full-face collection audited; no Bangladesh-wide fairness conclusion.
ChokePoint [R7]	Public surveillance-oriented face-set dataset and selection research.	Custom crop-based supplement, not the full official benchmark.
ICFD / D2 SC-GAN [R8]	Laboratory/classroom domain context.	Public sample engineering check only; no GAN trained or full release evaluated.

Do not label this compact review a systematic review or claim first-in-literature status. The previous report's unresolved citations, unknown-author rows and broad surveillance papers require individual verification if retained. The core references supplied at the end are sufficient to support the present method descriptions, not a claim of exhaustive literature coverage.

2.3 Gap Analysis

Frame a precise experimental gap without claiming that the individual methods are new.

Ready-to-adapt text.

A pretrained recognizer does not specify how a newly enrolled identity should be represented, when the closest gallery match should be refused, or whether a quality proxy improves multi-image pooling under a given input distribution. This project addresses those operating choices through paired enrollment comparisons, explicit rejection and failure accounting, and additional development and temporal sensitivity studies. The contribution is the declared protocol and resulting evidence, including negative results, rather than a new recognition backbone or a claim that template enrollment and quality weighting have not been studied before.

Research question	Evidence used	Limit
RQ1: Do a few original views improve acceptance?	Main final one/few-photo comparison	Frozen calibrated methods; public masked probes
RQ2: Do quality/variant choices help?	Development pooling and augmentation	Reused calibration/assessment; one declared variant recipe
RQ3: How does gallery composition matter?	Fixed-probe 10/25/55-ID growth	Small ten-ID known core; no arbitrary-scale guarantee
RQ4: Does temporal weighting beat averaging?	ChokePoint pooling and contamination	Synthetic effects; identities recur; no live tracker
RQ5: What are the tested software costs?	CPU stage/persistence/search timings	Warm fixed inputs; camera/UI/network excluded

2.4 Summary

Close with the methods selected from the reviewed background.

Ready-to-adapt text.

The review motivates a frozen embedding pipeline with explicit enrollment and rejection policies, while showing that these components have established prior work. The next chapter defines their implementation and experimental controls so that the results can be interpreted within the available data and deployment limits.

Chapter 3. Project Design

Ready-to-adapt text.

This chapter defines the functional and nonfunctional requirements, context, data flows and interface design. It then specifies the chosen methods, considered alternatives, experimental controls, project plan and task-allocation records.

3.1 Requirement Analysis

Describe who uses the research prototype and what an input/output means. A recognition decision is not permission to enter a building. Research operators, participants represented in the datasets, authors and future institutional administrators have different interests.

3.1.1 Functional and Nonfunctional Requirements

Use identifiers so tests and design choices can be traced to requirements.

ID	Requirement	Implemented evidence / limit
F1	Enroll an identity using one to three original photos.	Mean-template insertion; exactly one usable face per enrollment image in app.
F2	Search gallery and explicitly accept/refuse a probe.	Score/margin/quality engine and accepted-ID output.
F3	Persist, reload, update and remove templates.	JSON/NPZ gallery; isolated lifecycle checks.
F4	Keep queued unknown identities out of search until reviewed.	Pending AutoID filtering and approval functions; not liveness or identity proof.
F5	Present decisions, scores, margins and quality.	Streamlit still-image UI and defense cases.
F6	Record experiment outcomes including failures.	Saved per-probe tables, settings and verification records.
N1	Fail explicitly when real model assets are unavailable.	Synthetic fallback disabled; explicit test-only mock mode.
N2	Preserve paired samples and frozen evaluation settings.	Hash, identity-role and probe-overlap checks.
N3	Measure practical CPU costs.	Median/p95 image and gallery timings, not guaranteed camera FPS.
N4	Support modular maintenance and local prototype use.	Separate detection/alignment/recognition/gallery modules.
N5	Respect data handling and human review needs.	Portrait previews off by default and local benchmark data; production controls remain future work.

3.1.2 Context Diagram

Use Figure 3.1 below. The operator enters enrollment/probe data and receives decisions. The system uses a local persistent gallery and frozen model assets. Public research datasets are inputs for evaluation, not a live campus student database.

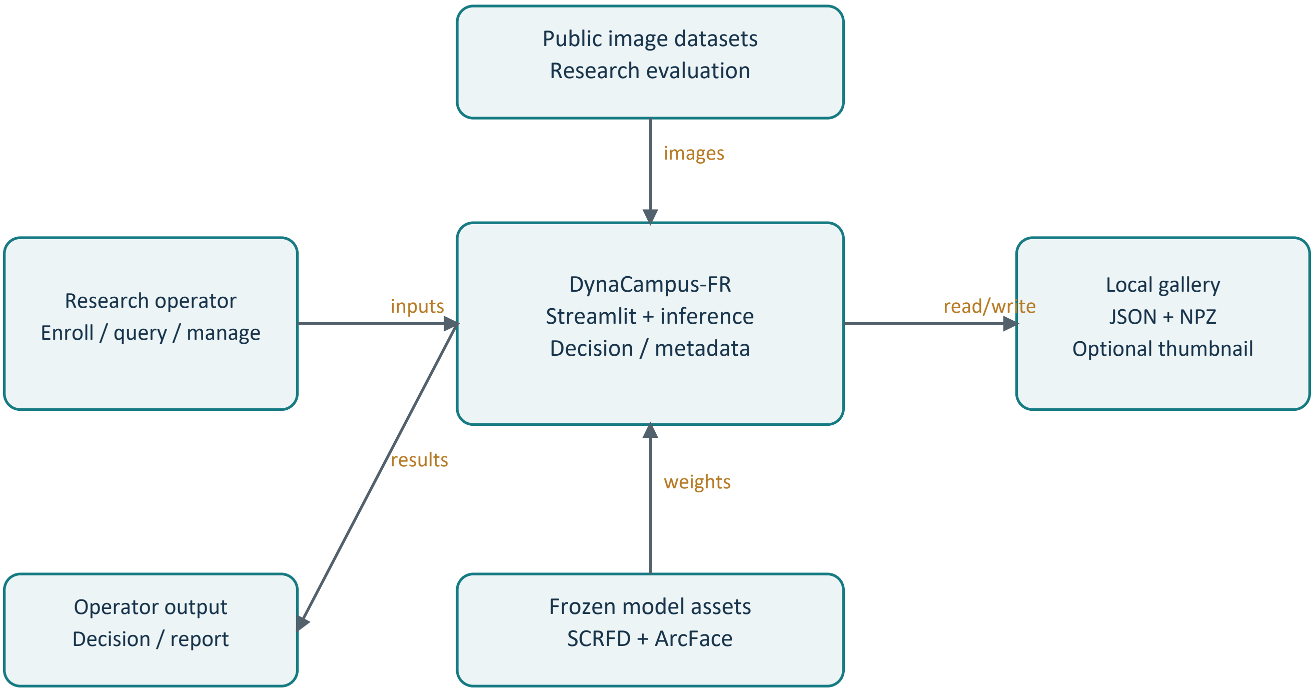

Public image datasets
Research evaluation
images
Research operator
Enroll / query / manage
inputs
DynaCampus-FR
Streamlit + inference
Decision / metadata
read/write
Local gallery
JSON + NPZ
Optional thumbnail
results
weights
Operator output
Decision / report
Frozen model assets
SCRFD + ArcFace

Figure 3.1. System context for the implemented research prototype.

3.1.3 Data Flow Diagram Level 1

Use Figure 3.2. Processes show image preparation, detection/alignment, embedding extraction and separate enrollment/recognition routes. The displayed gallery is a data store. Label arrows with data rather than implying training occurs at each insertion.

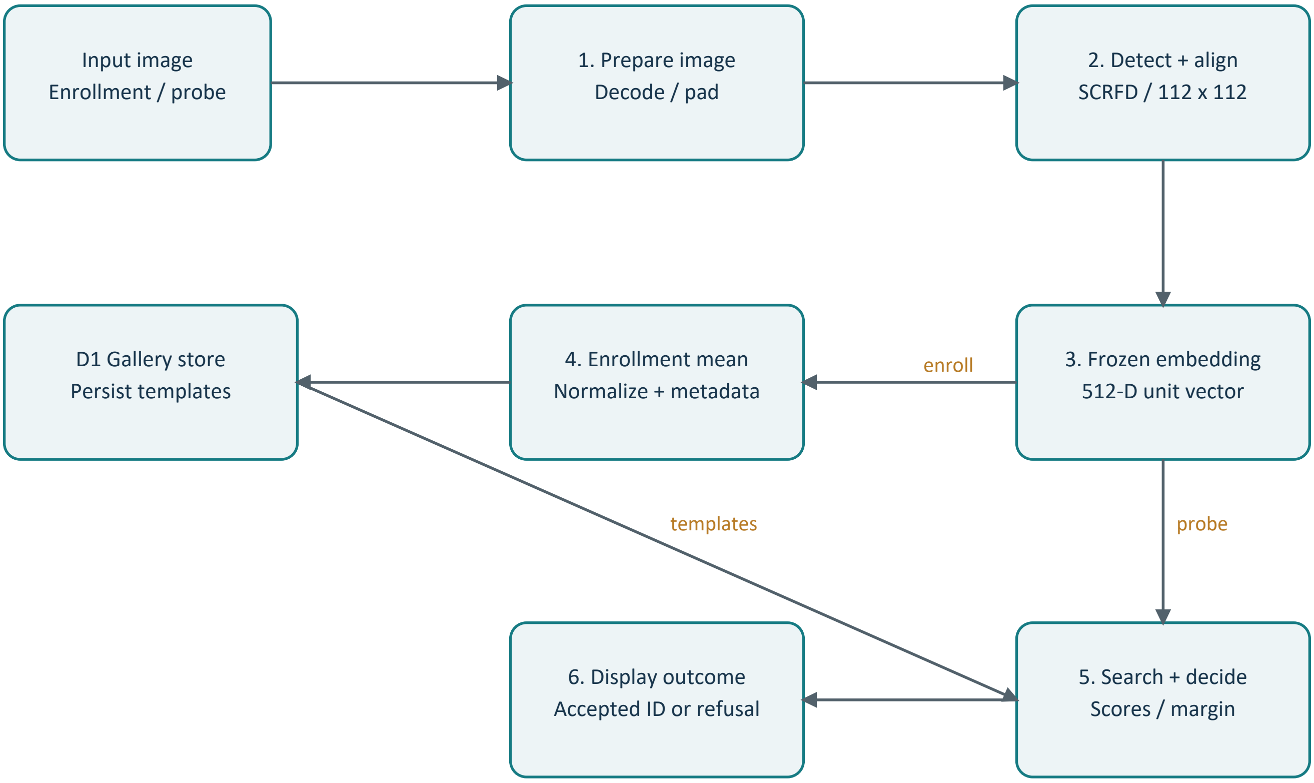

Input image
Enrollment / probe
1. Prepare image
Decode / pad
2. Detect + align
SCRFD / 112 x 112
3. Frozen embedding
512-D unit vector
enroll
4. Enrollment mean
Normalize + metadata
D1 Gallery store
Persist templates
templates
probe
5. Search + decide
Scores / margin
6. Display outcome
Accepted ID or refusal

Figure 3.2. Level-1 data flow; enrollment writes templates and recognition reads them.

3.1.4 UI Design

Provide the wireframe below and describe actual operator actions. A wireframe is a design figure, not a screenshot proving a live camera test.

- Enrollment: name/subject ID, up to three original photos, one-face validation, template confirmation and success/failure.

- Recognition: image upload, tight-crop toggle, frozen/manual settings, outcome and accepted identity only for KNOWN; show candidate scores without presenting them as confirmed identity.

- Gallery management: list entries, reload/persistence behavior, removal; optional queued unknown review.

- Evidence app: paired development examples and an isolated ten-step gallery walkthrough. Camera controls exist but are not operational-performance evidence.

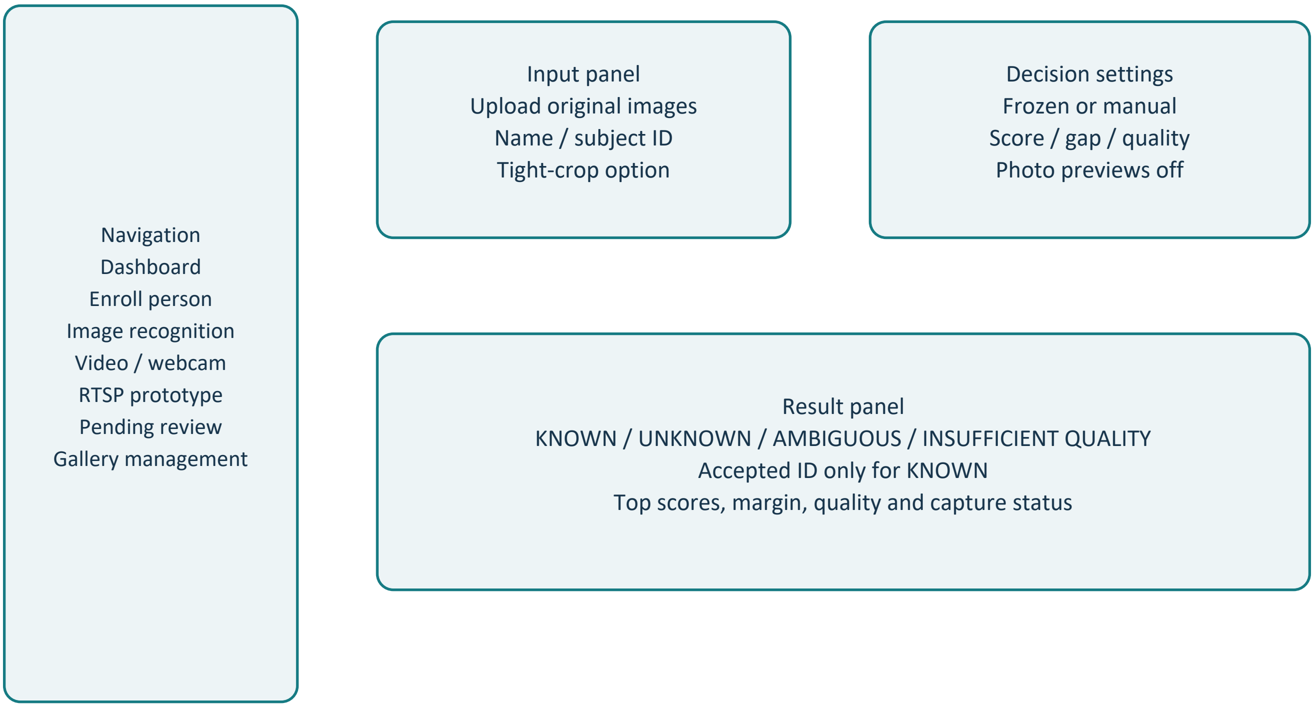

Navigation
Dashboard
Enroll person
Image recognition
Video / webcam
RTSP prototype
Pending review
Gallery management
Input panel
Upload original images
Name / subject ID
Tight-crop option
Decision settings
Frozen or manual
Score / gap / quality
Photo previews off
Result panel
KNOWN / UNKNOWN / AMBIGUOUS / INSUFFICIENT QUALITY
Accepted ID only for KNOWN
Top scores, margin, quality and capture status

Figure 3.3. UI wireframe for writing the interface-design section; no live performance is implied.

3.2 Detailed Methodology and Design

Organize this section using the following suggested 3.2.x subheadings. The original template does not prescribe these subheadings; this expansion preserves its required top-level structure. Include alternatives and rationale rather than only an implementation diagram.

Alternative considered	Reason for the selected choice	Trade-off / boundary
Train a seven-class CNN vs frozen embedding gallery	Frozen embeddings allow insertion without updating class outputs or retraining.	Pretraining provenance/overlap can limit causal and population claims.
Haar/edge pipeline vs SCRFD landmarks	Use a real detector and shared five-point alignment compatible with pretrained recognizers.	No isolated detector accuracy comparison is claimed.
One view vs mean of several originals	Test whether additional original views improve the enrollment template.	More acquisition effort; availability varies.
Mean vs quality-weighted/best selection	Mean is transparent; quality alternatives are evaluated instead of presumed superior.	Heuristic proxies may saturate or respond to noise.
Fixed score vs score and top-two gap	Explicitly test ambiguity in a gallery search.	Refusals reduce coverage; thresholds need representative calibration.
NumPy exact search vs indexed vector database	Small 55-ID galleries make exact search simple and inspectable.	No large-scale search benchmark or FAISS implementation.
Local JSON/NPZ vs SQL backend	Sufficient for an isolated prototype and persistence demo.	Concurrency, security, audit history and institutional integration are not supplied.
Synthetic adapter training vs no adapter in final method	Random-vector training cannot validate CCTV adaptation.	Real paired domain data and independent evaluation are future work.

3.2.1 Data sources, audit and original/derived separation

State these counts and exclusions exactly. Dataset names describe provenance; local usable counts describe your actual experiment.

Dataset / role	Local inputs	Interpretation
halfFace v 2 originals	941 files / 166 IDs initially; 926 images / 164 IDs usable.	12 files from conflicting duplicate IDs p039/p090 plus 3 ambiguous-condition files excluded.
halfFace derived collections	933 upper-face files and 3,741 augmented upper-face files remain archived.	Not independent new participants or the main evaluation input.
ICFD public sample	400 images / 10 IDs; 200 source-gallery and 200 source-probe images.	Five development IDs, five reserved; no unknown identity probes.
ChokePoint supplement	4,776 original crops inventoried in four selected sequences; 3,292 distinct source crops used for selected tracks/gallery.	Supplied 96 x 96 eye-aligned crops; one portal and custom protocol.

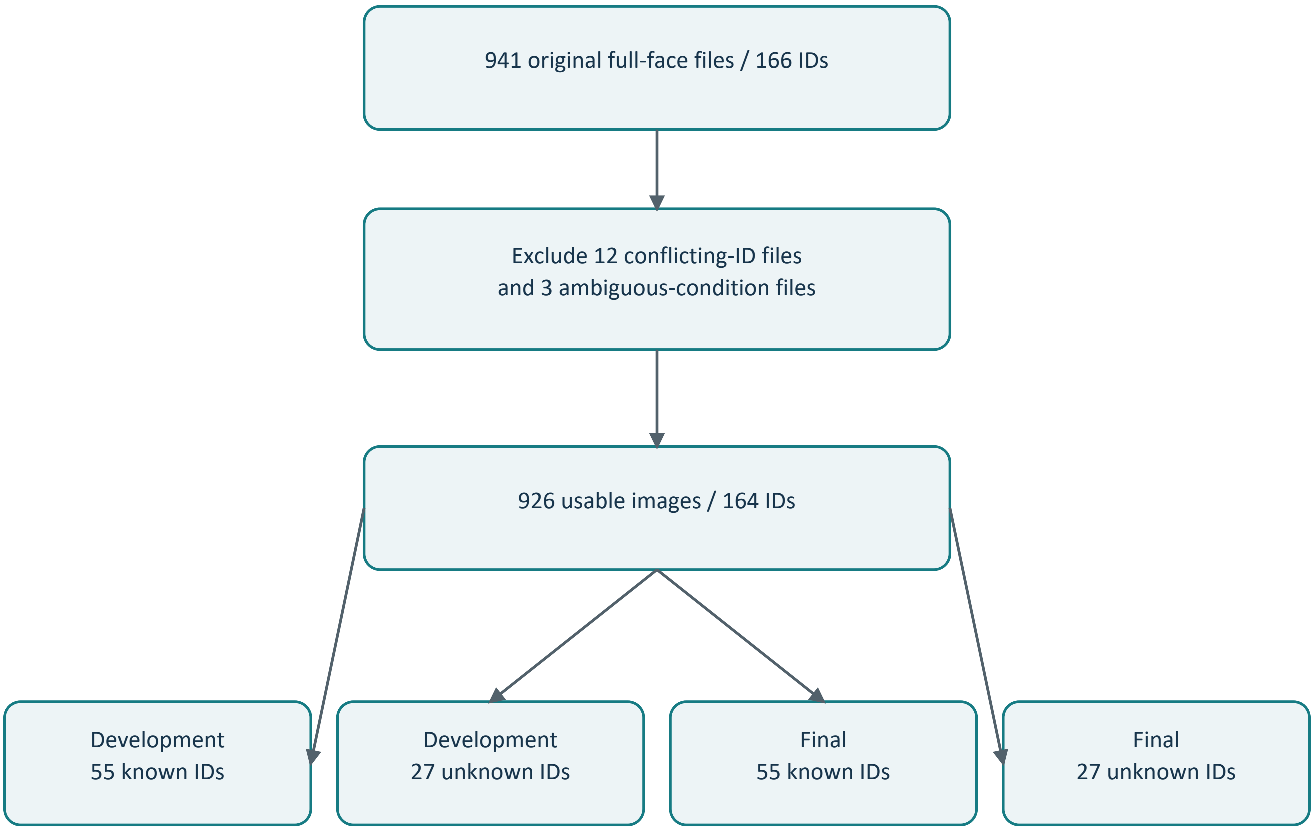

941 original full-face files / 166 IDs
Exclude 12 conflicting-ID files and 3 ambiguous-condition files
926 usable images / 164 IDs
Development 55 known IDs
Development 27 unknown IDs
Final 55 known IDs
Final 27 unknown IDs

Figure 3.4. Main dataset audit and frozen identity-role partition.

3.2.2 Sampling, roles and leakage controls

Unknown identities must be absent from their corresponding gallery. Known probes use original masked halfFace photos; enrollment uses original unmasked frontal/left/right photos. The one-photo method chooses the documented frontal candidate. Up to three means available usable originals, not three synthetic copies.

- Freeze 55/27 development and 55/27 final known/unknown identity roles before recognition. The main final test contains 160 known and 151 unknown probe attempts.
- Reject enrollment/probe path reuse and exact duplicate hashes across development/final evaluation inputs. Keep role selection independent of a favorable observed match.

- Record no-face, multiple-face, unreadable and invalid-landmark outcomes; do not replace failed inputs with random embeddings or silently drop difficult known attempts.
- Identity-disjoint here means the local experiment roles. It does not establish that a public pretrained checkpoint has never seen these people or near-duplicate imagery.

3.2.3 Detection, alignment and frozen recognizers

For tight benchmark crops, apply a constant neutral border of half the source height/width on each side with value 127. Detect at confidence 0.50 using SCRFD640 x 640, require exactly one usable face in the research extraction, and use five-point similarity alignment to 112 x 112.

- ArcFace accepts aligned BGR input through the InsightFace ONNX wrapper, which performs its own channel handling and normalization. Do not add an unverified second channel swap.
- AdaFace IR-50/MS1 MV2 accepts uint 8 BGR; applies (pixel/255 -0.5)/0.5 and returns a unit vector plus raw feature norm. Its preprocessing was checked against pinned source.
- ArcFace and AdaFace share the same aligned crops in the development comparison. They use different pretraining data and ONNX/PyTorch CPU engines; observed differences do not isolate training loss.
- The main dashboard uses ArcFace. AdaFace is a separate research comparison. FaceLiVT is an unfinished integration, not a completed result.

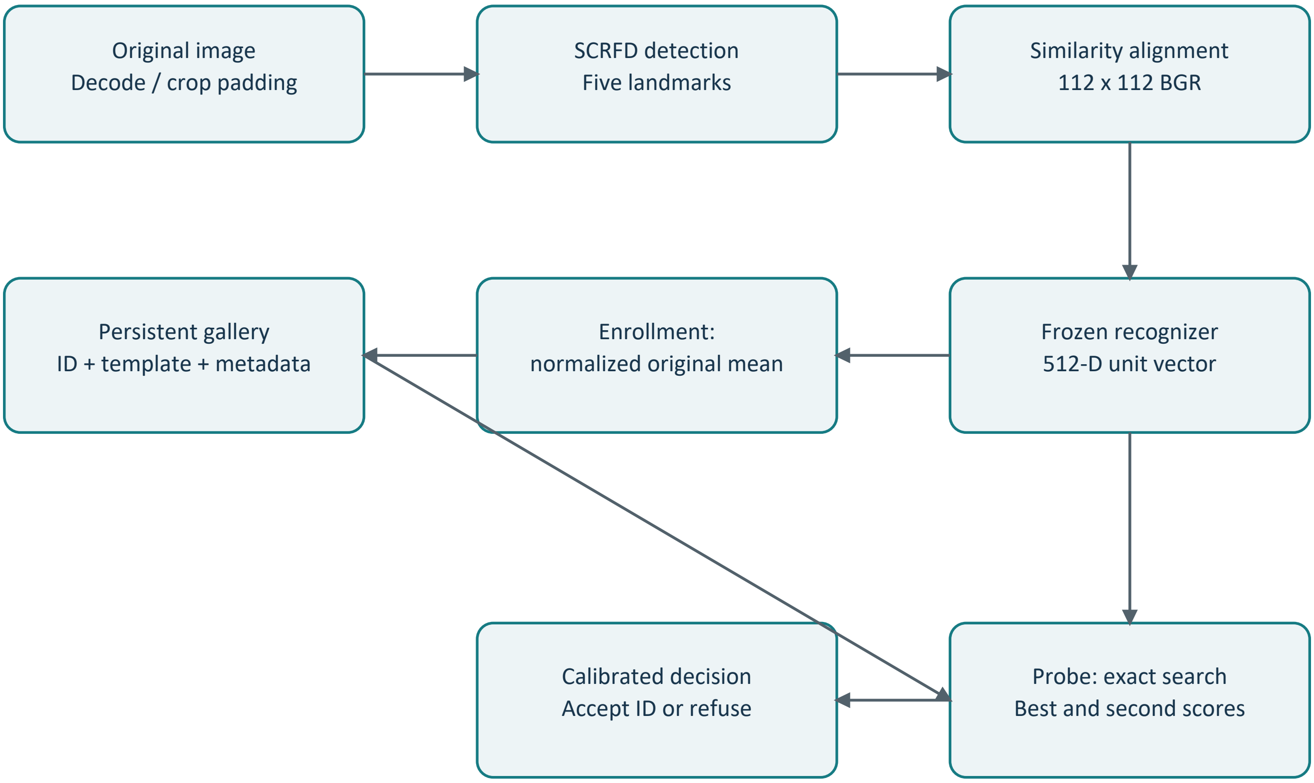

Original image
Decode / crop padding
SCRFD detection
Five landmarks
Similarity alignment
112 x 112 BGR
Persistent gallery
ID + template + metadata
Enrollment:
normalized original mean
Frozen recognizer
512-D unit vector
Calibrated decision
Accept ID or refuse
Probe: exact search
Best and second scores

Figure 3.5. Implemented still-image inference and template routes.

3.2.4 Embeddings, templates and similarity equations

Define x as an aligned crop, f as a frozen recognizer, z as a unit embedding, g_j as identity j's template, and s_j as cosine similarity. Use the following equations in the thesis; vectors are 512-dimensional.

$$z_i = f(x_i) / ||f(x_i)||_2$$

$$g_j = \mathrm{mean}(z_{j1}, ..., z_{jn}) / ||\mathrm{mean}(z_{j1}, ..., z_{jn})||_2$$

$$s_j = g_j^T z;\ s1 = \max_j(s_j);\ gap = s1 - s2$$

Weighted ablation: $g_j = \mathrm{normalize}(\mathrm{sum}_i(w_i z_{ji}) / \mathrm{sum}_i(w_i))$

For one-photo enrollment $n=1$. For few-photo originals n is at most 3 and source failures are recorded. Reject non-finite/zero vectors; normalize the final mean. The default research template is an unweighted mean. Scores lie in the cosine range, not in a calibrated probability scale. Pending unapproved AutoID entries are excluded from searchable identities.

3.2.5 Four-state open-set rule and capture failure

Let tau be score threshold, m be minimum top-two gap, q be the heuristic quality and q_{min} the quality gate. A failure to obtain an embedding is an upstream outcome, not an unknown-person decision. If the searchable gallery has one identity, the second-score ambiguity test is handled separately.

```
if extraction fails: CAPTURE_FAILURE (outside decision engine)
else if q < q_min: INSUFFICIENT_QUALITY
else if no candidate or s1 < tau: UNKNOWN
else if at least two candidates and s1 - s2 < m: AMBIGUOUS
else: KNOWN, accepted identity = best gallery ID
```

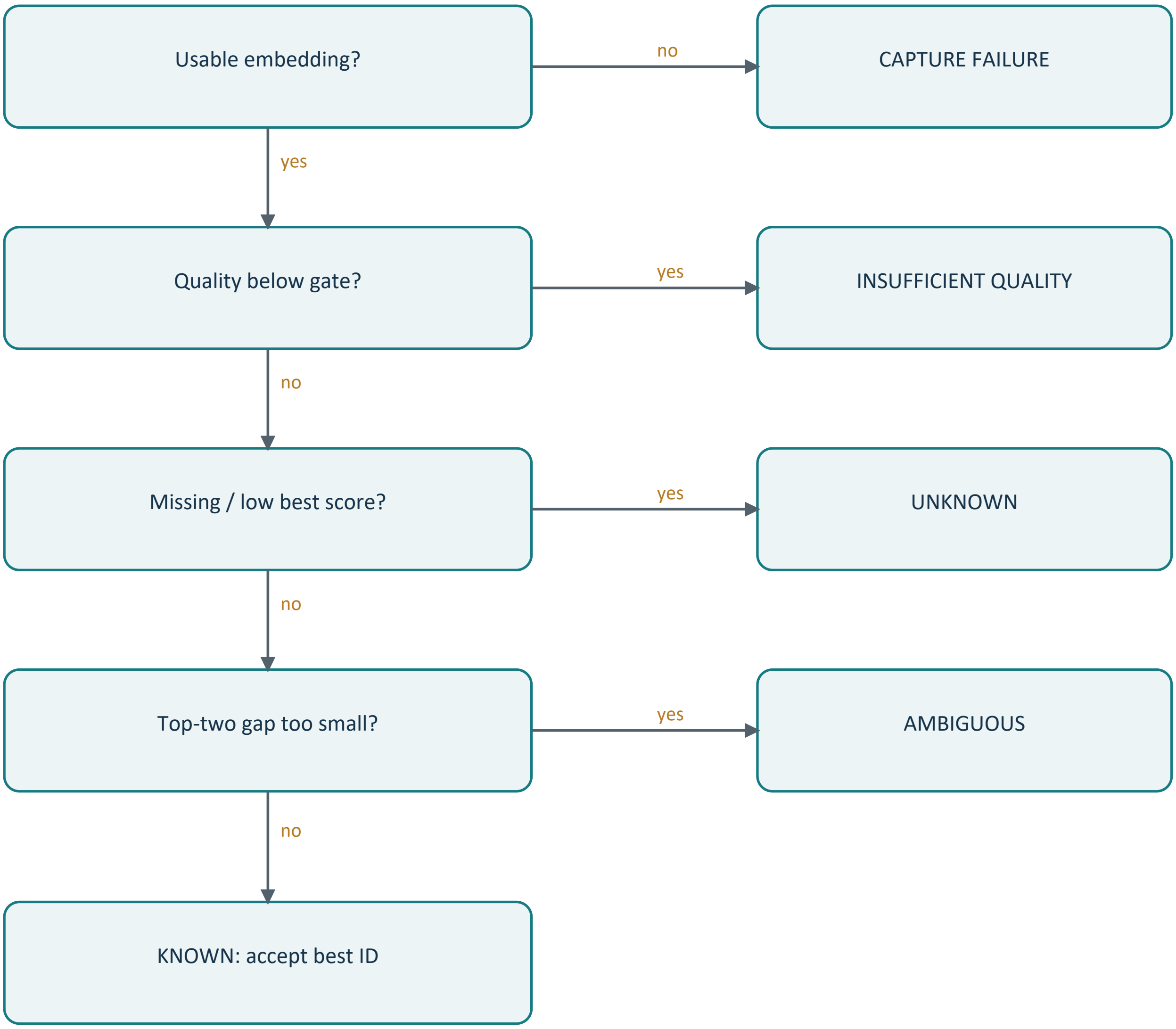

Usable embedding?
no
CAPTURE FAILURE
yes
Quality below gate?
yes
INSUFFICIENT QUALITY
no
Missing / low best score?
yes
UNKNOWN
no
Top-two gap too small?
yes
AMBIGUOUS
no
KNOWN: accept best ID

Figure 3.6. Decision sequence. A low-quality input never receives a relaxed main match threshold.

3.2.6 Calibration, selection and final freeze

The main development grid maximizes correct known acceptance while conditional unknown FPIR is at most 1% and valid decision coverage is at least 85% separately for known/unknown probes. Coverage here counts KNOWN or UNKNOWN, not ambiguity or quality refusal.

- Candidate score thresholds use observed development scores and next-representable boundaries so the >= acceptance convention and score ties are handled explicitly.
- Margin grid: 0,.025,.05,.075,.10,.15,.20. Quality-family grid: 0,.20,.30,.40,.50,.60,.70,.80,.90; other families disable quality.
- Tie-break in order: maximize correct known accepts; minimize false unknown accepts; minimize wrong known identities; maximize coverage; prefer lower quality and margin gates; then higher score threshold.
- Main primary few-photo method: tau=0.3778589367866516, m=.15, q_min=0. Freeze this method, source/configuration hashes, model hashes, split and sample manifest before final inference.
- One-photo and few-photo score-only thresholds are separately calibrated. The final comparison has no test retuning, degradation or backbone training. An exclusive start receipt prevents accidental repeat inference.

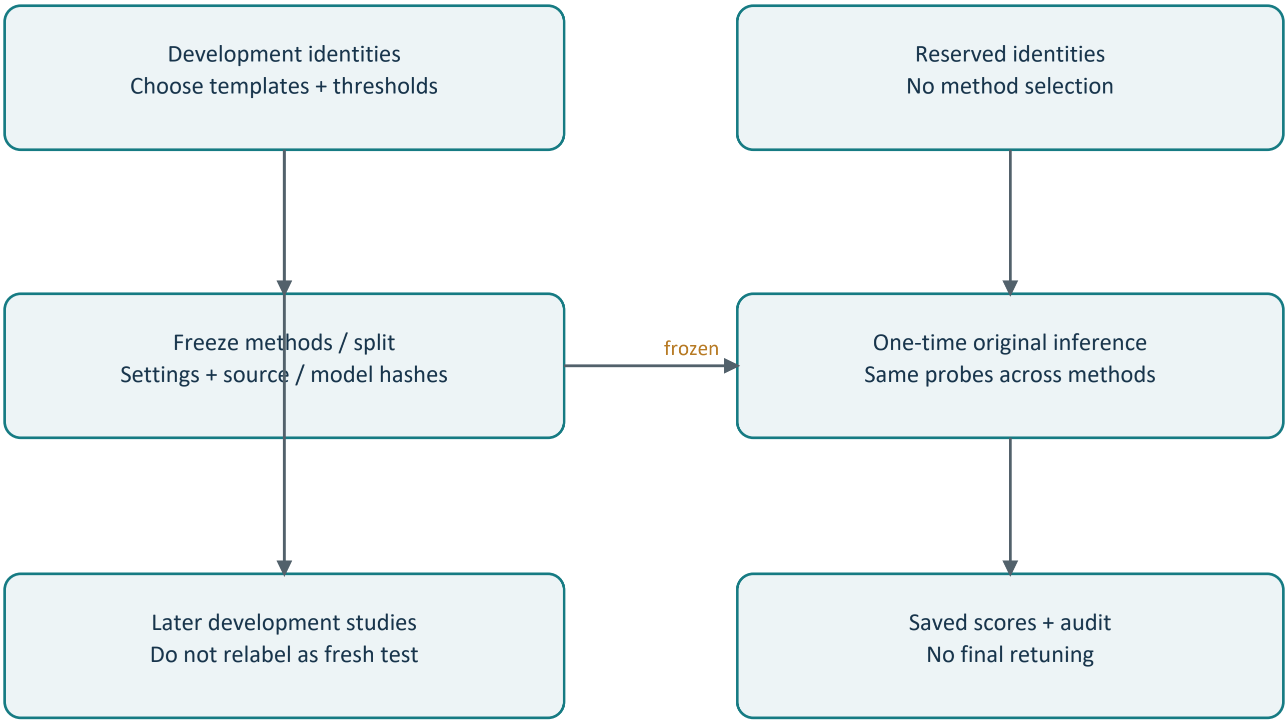

Development identities
Choose templates + thresholds
Reserved identities
No method selection
Freeze methods / split
Settings + source / model hashes
frozen
One-time original inference
Same probes across methods
Later development studies
Do not relabel as fresh test
Saved scores + audit
No final retuning

Figure 3.7. Development selection and one-time reserved evaluation. Later studies remain separate.

3.2.7 Quality assessment and original-photo pooling

The heuristic quality is .45*blur+.35*resolution+.20*pose. Blur uses 1-exp(-Laplacian variance/100); resolution maps the native shorter face-box dimension from 20 to 112 pixels into[0,1]; pose uses eye/nose symmetry and eye tilt. These are engineering proxies, not learned demographic or identity-quality estimates. The final selected gate is 0, so it refuses no valid probe for quality.

- Pooling ablations use the same original candidates: uniform mean, normalized heuristic weights, normalized detector-confidence weights and the actual highest-quality original.

- The pooling experiment uses a recalibrated score-only view and a fixed previous uniform-mean score/margin sensitivity view. Do not confuse this fixed one-photo result with a separately calibrated backbone-study one-photo margin result.

- Linear normalized weights in the original live buffer are not softmax, despite an old comment. Do not label them a novel trained QATA network.

3.2.8 Synthetic enrollment and fixed-probe gallery composition

Augment only the aligned frontal enrollment crop. The original and five variants each receive 1/6 weight before mean normalization. Probes remain unchanged. Variants: Gaussian 3 x 3 sigma 1; cubic down/up 56; JPEG 50; gamma 0.7 and 1.4 using(pixel/255)^gamma. Redetection is not run on variant crops.

- Compare one original, one plus variants and up to 3 originals for both checkpoints and both rules. Retain both fixed-one-photo and separately recalibrated development views.

- Growth uses the first 10 IDs in the frozen development-known order as a fixed core; 28 known and 150 unknown attempts stay identical. Other known IDs only add background gallery entries.

- Evaluate one 10-ID gallery, five 25-ID compositions with seeds 42,123,2026,7,99, and one 55-ID gallery. Keep previous 55-ID thresholds fixed. Composition repeats are not independent samples or confidence intervals.

3.2.9 Controlled robustness and temporal supplement

Keep the two corruption experiments distinct. Image robustness acts on original-size development probes before detection; the temporal study corrupts supplied aligned crops after resizing. Their severity labels are not interchangeable.

Experiment	Transform recipe	Controlled boundary
Image robustness	Blur sigma .003/.006/.012 of original short edge; long-edge downsampling cap 128/64/32 with INTER_AREA and no upscaling; JPEG 70/30/10.	Original plus 9 single-factor conditions; fixed gallery/thresholds; no combined profiles.
ChokePoint mild	Gaussian 3 x 3 sigma 1; JPEG 75.	All crops first cubic-resized 96 to 112.
ChokePoint medium	Gaussian 5 x 5 sigma 2; cubic 112 to 56 to 112; JPEG 50.	Same source pixels across policies.
ChokePoint severe	Gaussian 7 x 7 sigma 3.5; cubic 28; independent normal noise sigma 15, round/clip; JPEG 25.	Noise can favor averaging; not a model of every real camera.
ChokePoint mixed	Per-frame clean/mild/medium/severe probabilities .2/.3/.3/.2.	Stable SHA256-derived path/condition seeds, not Python randomized hash().

ChokePoint uses 17 gallery-known and 8 unknown IDs. Gallery: P1E_S1_C1, first available numeric crop per known ID; no frontal/optimal claim. Calibration: P1E_S2_C2, 143 known and 66 unknown tracks. Evaluation: P1E_S3_C3 and P1E_S4_C1, 303 known and 143 unknown tracks. Every query has 5 consecutive non-overlapping frames, with at most 15 eligible tracks per ID/sequence. Identity populations recur between calibration/evaluation; the pilot already saw P1 E.

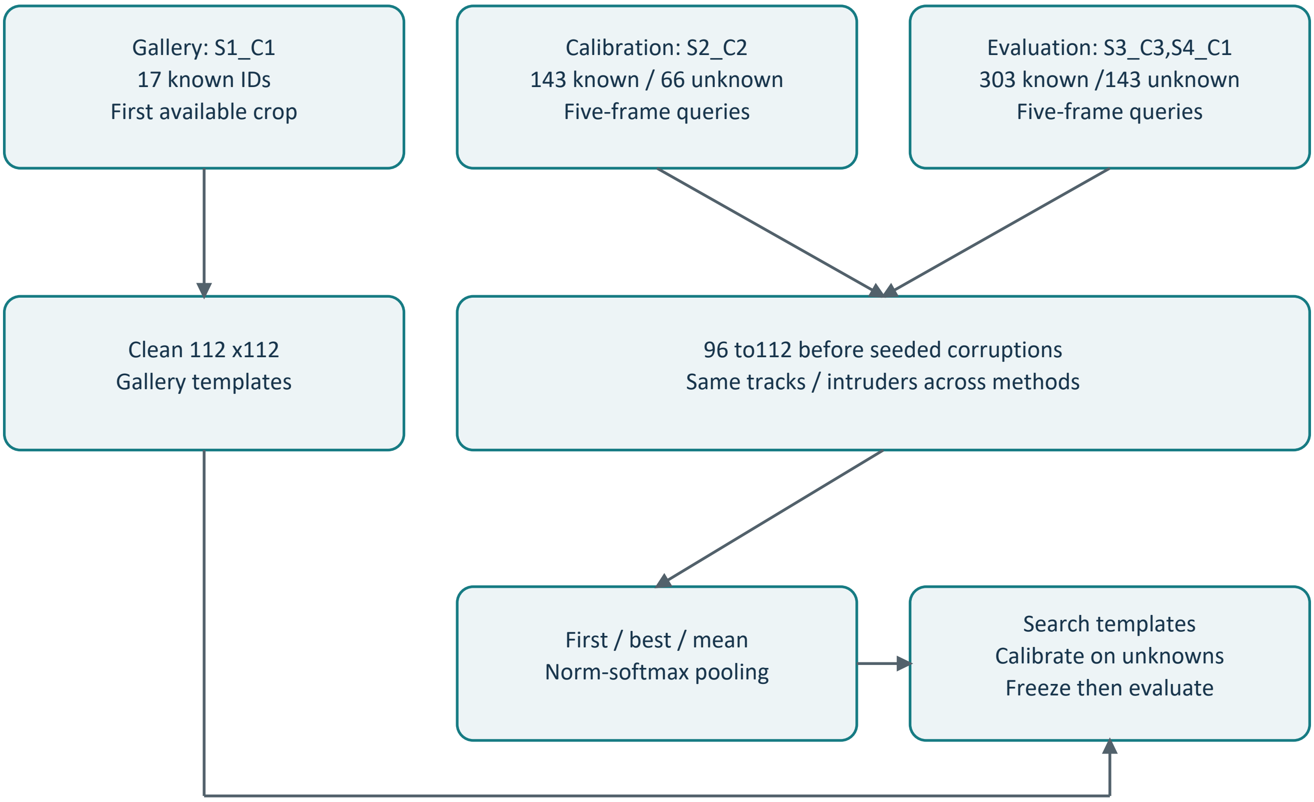

Gallery: S1_C1
17 known IDs
First available crop
Calibration: S2_C2
143 known / 66 unknown
Five-frame queries
Evaluation: S3_C3,S4_C1
303 known /143 unknown
Five-frame queries
Clean 112 x112
Gallery templates
96 to112 before seeded corruptions
Same tracks / intruders across methods
First / best / mean
Norm-softmax pooling
Search templates
Calibrate on unknowns
Freeze then evaluate

Figure 3.8. Custom ChokePoint supplementary protocol. Supplied crops bypass detector and live tracker.

Temporal policies are first frame, seeded random, best Laplacian/contrast, best raw feature norm, mean, and standardized norm-softmax beta 0.5,1,2,5. Retain raw pre-normalization norms. Simulated contamination substitutes one foreign-identity last frame per known track, reusing the same intruder across policies and matched-tier/mild-tier controls. It is not measured tracker error.

Mean temporal: z_track = normalize(mean_t(z_t))

a_t = beta * (norm_t - mean(norm)) / (std(norm) + 1e-8)

w_t = exp(a_t - max(a)) / sum_u(exp(a_u - max(a)))

Norm-softmax track = normalize(sum_t(w_t * z_t))

The supplementary AOSM diagnostic differs from the main four-state engine: it accepts if score >= max(tau-(1-q)*.08,.20). Track quality averages clipped(norm-16)/6 values. Calibrate each policy/condition/rule using unknown calibration maximum scores, taking the lowest threshold with at mostfloor(.01*66)=0 false accepts. Do not use evaluation labels for that selection. Boundaries 16-22 originate in the examined pilot, so quality-proxy validation is exploratory.

3.2.10 Storage, persistence, runtime and implementation boundary

Store subject metadata in JSON, templates in NPZ and optional thumbnails as JPEG. There is no implemented PostgreSQL database or production recognition-event history. An event JSONL log in an isolated defense demonstration is a separate artifact, not a campus audit service.

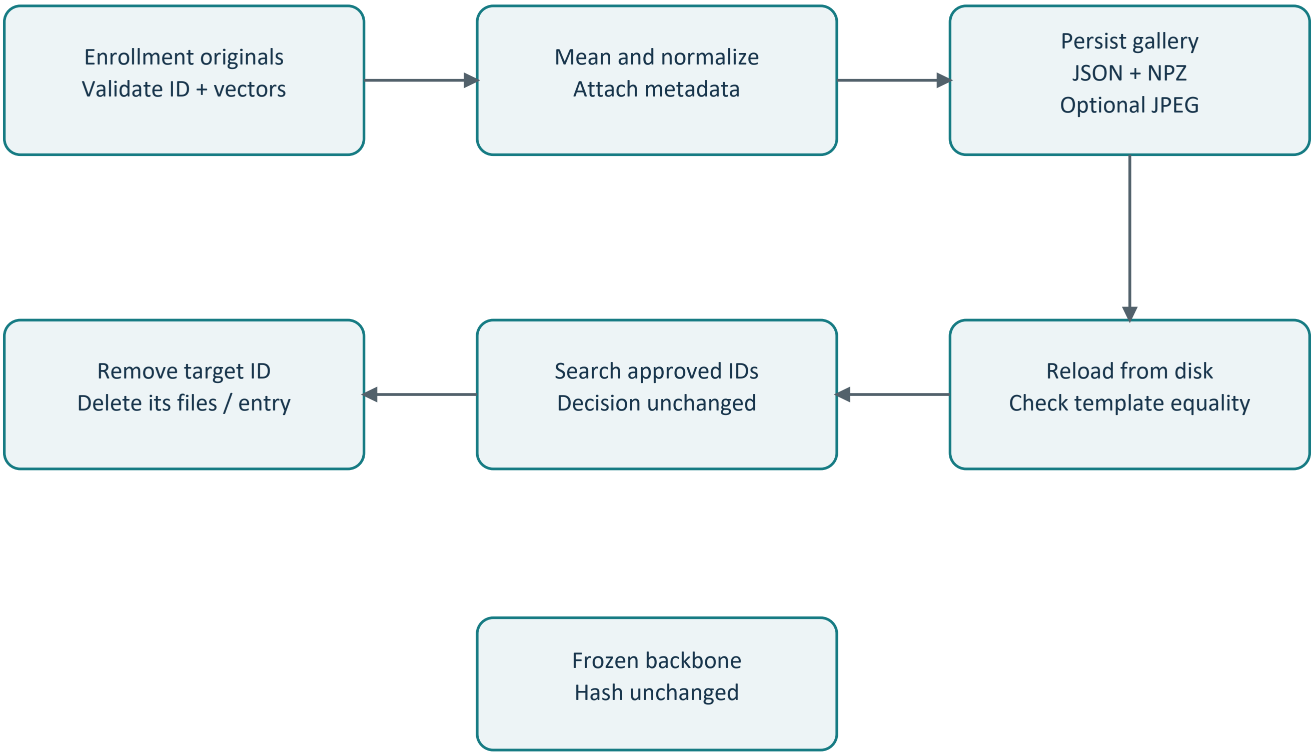

Enrollment originals
Validate ID + vectors
Mean and normalize
Attach metadata
Persist gallery
JSON + NPZ
Optional JPEG
Reload from disk
Check template equality
Search approved IDs
Decision unchanged
Remove target ID
Delete its files / entry
Frozen backbone
Hash unchanged

Figure 3.9. Gallery lifecycle and implemented storage. Backbone weights remain unchanged.

Runtime measures 30 enrollment and 60 recognition image calls per checkpoint. Enrollment includes decoding through a fresh persistent 55-to 56-ID insertion; restoring the benchmark gallery occurs outside the timed window. Recognition measures decoding through actual search and decision. Separate 300-call search/decision repetitions at 10/25/55 IDs include normalizing, rebuilding the gallery matrix and sorting. Exclude model loading, warmup, UI/browser transport, capture and RTSP network time. Use direct wall times, not sums of stage medians.

3.2.11 Metrics and uncertainty

Define these denominators before presenting any percentages. Failure and refusal outcomes are not silently counted as correct rejection.

Metric	Definition / denominator	Reporting rule
End-to-end Rank-1	Correct top identity / all known probe attempts.	Includes failed extraction attempts in denominator.
Correct known acceptance	Correct identity AND decision KNOWN / all known attempts.	Main headline; more demanding than retrieval.
Conditional unknown FPIR	Unknown probes accepted KNOWN / unknown probes with usable embeddings.	Includes ambiguous/quality-refused valid unknowns in denominator; do not restrict to covered cases.
End-to-end unknown acceptance	Unknown KNOWN decisions / all unknown attempts.	Different denominator when capture failures exist.
Decision coverage	KNOWN or UNKNOWN / all attempts, or / valid extractions.	Always label end-to-end or valid; AMBIGUOUS/quality refusals excluded.
Wrong known identity accept	Known probe accepted as another gallery ID.	Separate from rejection and unknown false acceptance.
Pairwise TAR / FAR / EER	Compare genuine and impostor pairs; operating points from pair score distribution.	Not equal to max-gallery unknown-query FPIR.
Latency median / p95	50 th/95 th percentile of stated wall-time samples.	Report calls, CPU/thread settings and excluded stages.

For paired effects, resample known identities with replacement, retaining all their probes and pairing policies by identical source path. Main and development intervals use 2,000 resamples, seed 42; ChokePoint uses 1,000 identity resamples. Report 2.5 th/97.5 th percentiles. Intervals condition on fixed calibration; they do not integrate calibration uncertainty, unseen sites or unknown-identity population uncertainty. Multiple exploratory comparisons do not support unadjusted blanket significance claims. An all-zero false-accept bootstrap interval must not be read as a population zero-risk bound.

3.3 Project Plan

Present a chronological process and dependencies; obtain exact historical dates from the team instead of inventing a Gantt chart. The recorded run identifiers establish execution snapshots, not every member's work schedule.

Phase	Required output	Completion state
Scope and literature	Research questions, selected components and contribution boundary	Available; references and final wording need editorial review.
Data audit	Original/derived inventory, exclusion log and frozen roles	Completed for reported public inputs.
Development	Baseline, enrollment/decision selection	Completed; later descriptive ablations also available.
Freeze and final evaluation	Saved primary methods and one-time original-image results	Completed; preserve settings.
Supplementary analysis	Backbones, pooling, stress, variants, growth, temporal and runtime	Reported evidence available with separate scopes.
Software/demo verification	Tests, result recomputation and isolated gallery lifecycle	Available; representative camera benchmark not completed.
Writing and submission	Template-complete chapters, captions, references and supervisor review	This guide supplies content; actual final submission remains team work.

3.4 Task Allocation

Keep the actual five-person team table, but fill contribution claims from real records. The following allocation is a proposed writing plan, not a statement of historical authorship.

Writing role	Deliverable	Cross-check partner
Research editor	Title, abstract, Chapters 1/6, terminology and final integration	Results author checks every headline.
Background author	Chapter 2, primary references and gap/application tables	Methods author checks relevance and novelty wording.
Methods/design author	Chapter 3, equations, diagrams and alternatives	Software/results author checks code consistency.
Results author	Chapter 4, numeric tables, uncertainty and software-test descriptions	Another member independently checks denominators.
Engineering/format editor	Chapter 5 mappings, budgets, lists, layout and submission checks	Whole team confirms real costs and roles.

For the actual contribution table, include member name/ID, implementation or analysis task, evidence produced, writing task, review task and confirmed date/status. Do not award every member all tasks or invent member percentages. All authors must review the final claims and acknowledge outside assistance according to university policy.

3.5 Summary

The original template requires this summary; it was absent from the old report's design chapter.

Ready-to-adapt text.

The design combines frozen recognition components with persistent mean templates and explicit open-set refusal rules. Declared data roles, calibration procedures, alternative-policy comparisons and failure accounting determine the experimental boundaries. The next chapter reports implementation checks and results under those boundaries, keeping the final comparison separate from development and exploratory evidence.

Chapter 4. Implementation and Results

Ready-to-adapt text.

This chapter describes the implementation environment, software and protocol checks, and the measured results. The predeclared final halfFace comparison is reported first, followed by descriptive development studies, an exploratory temporal supplement and engineering measurements.

4.1 Environment Setup

Report the environment actually used, not generic minimum hardware or a GPU model inferred from installed CUDA packages. The source record identifies the CPU family rather than a retail processor name; RAM capacity was not recorded for the timed run and must not be invented.

Component	Recorded configuration
Operating system	Windows-11-10.0.26200-SP0
Python	3.13.3
Processor	AMD64 Family 23 Model 113 Stepping 0, AuthenticAMD
Logical CPUs / inference threads	12 / 2
ArcFace engine	ONNX Runtime 1.27.0; CPUExecutionProvider
AdaFace engine	PyTorch 2.9.0+cu 126; CPU float 32
NumPy / OpenCV	2.3.3 / 5.0.0
CUDA use	False; CUDA-capable PyTorch installation does not mean GPU inference.
Interface / scientific tools	Streamlit, OpenCV, NumPy, pandas, SciPy, scikit-learn, Matplotlib, PyYAML and FilterPy.
Dependency snapshot	Recorded runtime declarations include Streamlit 1.65.0, InsightFace 1.0.1 and pytest 9.1.1; runtime/source versions can differ in formatting.
RAM / power / camera hardware	Not established for the recorded timing experiment. No camera is part of the measured latency boundary.

Model checkpoints are external pretrained assets, not products trained in this project. ArcFace uses the buffalo_l w 600 k_r 50 ONNX checkpoint and SCRFD det_10 g ONNX. AdaFace uses released IR-50/MS1 MV2 with source commit c60eaa786a42c03444f3df7096dbaf9d57ae010d. The appendix records hashes. Local availability and passing tests do not certify clean-machine installation or production deployment.

4.2 Testing and Evaluation

Use subheadings to separate software tests, protocol checks, empirical evaluation and engineering demonstrations. The top-level heading remains exactly the original template's heading. Do not present a software unit-test pass rate as recognition accuracy.

4.2.1 Software tests and behaviors verified

Fresh check for this guide: **104 tests passed, two upstream warnings, 8.60 seconds, successful process exit**. The command used the project virtual environment with pytest -q -p no: cacheprovider. The warnings concerned an upstream tensor-construction efficiency message and an upstream alignment API deprecation. Neither was a failed test. Root scratch transformer files are outside pytest's configured tests/ collection; the passing suite does not validate FaceLiVT.

Test module	Tests	What it checks / does not establish
test_adaface.py	6	Official preprocessing equivalence; invalid inputs/assets; restricted metadata checks.
test_adapter.py	1	MLP forward shape/normalization only; not domain adaptation quality.
test_aggregation.py	2	Buffer and linear weighted pooling behavior; not video accuracy.
test_augmentation.py	1	Earlier augmentation utility output behavior.
test_backbone_comparison.py	3	Paired probes, missing gallery identity handling and mean-template scoring.
test_baseline_protocol.py	4	Calibration ties, small unknown budgets and reserved-ID exclusion.
test_chokepoint_protocol.py	9	Stable seeds, sequence/gap rules, floor, calibration budget, pooling and paired statistics.
test_decision_ablation.py	8	Quality/score/gap ordering, equality boundaries, ambiguity and denominator accounting.
test_defense_cases.py	5	Retain harmed/failure cases; separate thresholds and paired source checks.
test_detection_alignment.py	4	Detector setup/empty input and synthetic landmark alignment checks.
test_dynamic_gallery.py	8	Insert/search/reload/remove, mean template, pending review, vectors and IDs.
test_embedder_safety.py	6	No silent fake fallback; explicit mock mode; model/input failures.
test_enrollment_comparison.py	5	Original candidate restrictions, overlap checks, paired probes and bootstrap.
test_enrollment_extensions.py	6	Deterministic variants, fixed core/probes, development guards and failure accounting.
test_enrollment_pooling.py	6	Normalized weights, fallback behavior, actual-sample selection and reserved-ID guards.
test_final_protocol.py	7	Frozen hashes/roles, no development overlap, manifest fields and repeat-inference guard.
test_open_set.py	3	Basic KNOWN/UNKNOWN/quality-refusal engine outcomes.
test_quality.py	6	Heuristic scores and visualization behavior; not learned quality validity.
test_recognition.py	6	Explicit mock embedding shape, normalization and cosine mathematics.
test_robustness_protocol.py	5	Deterministic transforms, sizing and failed-attempt denominators.
test_tracking.py	3	IoU/association and empty-frame behavior; not tracker ID accuracy.

4.2.2 Result verification and evidence scope

Check	Recorded verification	Interpretation
Final cache audit - rerun for this guide	1,133 prediction attempts; 1,126 searches; 585 source images; 0 development-ID overlap; frozen settings intact.	Reconstructs saved searches/decisions; no new reserved-image inference.
Backbone cache audit - rerun for this guide	2,832 attempts; 2,776 searches; 19,072 calibration entries; 571 shared images; ArcFace cache delta 0.	Shared detector outcomes and unchanged source inputs; development scope.
Pooling - existing receipt	3,540 attempts; 3,470 searches; 15 summaries; 808 reproduced baseline attempts.	Template/weight/decision checks; not new independent people.
Enrollment extensions - existing receipt	550 historical real-variant forwards; 12,280 predictions; 12,056 searches; 28,608 grid entries; 52 summaries.	Historical replay is not claimed to have been rerun while writing this guide.
Extension runtime - existing receipt	1,800 searches; 180 image calls; 5,220 stage samples; 6 persistent benchmark galleries.	Timing summaries/persistence checked; not camera FPS.
ChokePoint - existing receipt	41,265 predictions; 701,505 scores; 13,168 corrupted pixel arrays; 4,776 original crops; 42 paired intervals; 8 historical re-inference samples.	Custom exploratory experiment; existing checks not external peer review.
Defense - existing receipt	12 disclosed cases; 10 real backend steps; 10 AppTest steps; fresh features agree with cache; real multi-identity ambiguity.	Fresh browser capture=false; live screen recording=false.

4.2.3 Main and supporting experiment design

Study	Fixed comparison / unit	Calibration and assessment
Main final	Same 160 known/151 unknown originals; 55 gallery IDs; one/few-photo methods.	Development-selected settings frozen before reserved inference; 2,000 known-ID bootstrap draws.
Backbone and decision studies	Same 154 known/150 unknown development attempts; shared detector/crops.	Calibration and assessment reuse development data; descriptive.
Pooling / variants	Same development probes and original candidates; one declared 5-variant recipe.	Show recalibrated and fixed-rule views; pair by source path.
Robustness	Same 304 source probes; original+9 single-factor corruptions.	Original gallery/thresholds fixed; successful-extraction counts may change.
Growth	Same 28 known/150 unknown attempts; 10,25,55-ID galleries.	Existing 55-ID thresholds fixed; five 25-ID backgrounds, not independent cohorts.
ChokePoint	5-frame queries; 303 known/143 unknown evaluation tracks.	66 unknown calibration tracks per rule; separate sequences but repeated identities and pilot exposure.
Runtime/demo	Fixed real development images and isolated galleries.	Repeated wall-time calls measure execution, not recognition generalization.

For the final thesis, put the full sampling/calibration formulas in Chapter 3 and use this section to establish the experimental configurations, planned comparisons and measurement definitions before displaying outcomes. Do not report all tests as if they evaluate the same scientific question.

4.3 Results and Discussion

Organize the following as numbered 4.3.x subsections. Lead with the frozen final experiment; keep every later table labeled DEVELOPMENT, EXPLORATORY or ENGINEERING. Tables below are generated from saved numeric evidence, not transcribed from plot labels.

4.3.1 FINAL: one-photo versus few-photo enrollment

Table 4.1. Frozen final halfFace comparison.55 known gallery and 27 unknown identities; all known attempts retained.

Method	Gallery photos	Correct known /160	Unknown FA /151	Known acceptance 95% interval
One photo / score	55	130/160 (81.25%)	1/151	73.58% to 88.54%
Few photos / score	159	150/160 (93.75%)	0/151	88.48% to 98.11%
Few photos / score+gap	159	151/160 (94.38%)	0/151	88.89% to 98.74%

Table 4.2. Full-precision frozen operating settings. Thresholds differ across calibrated methods.

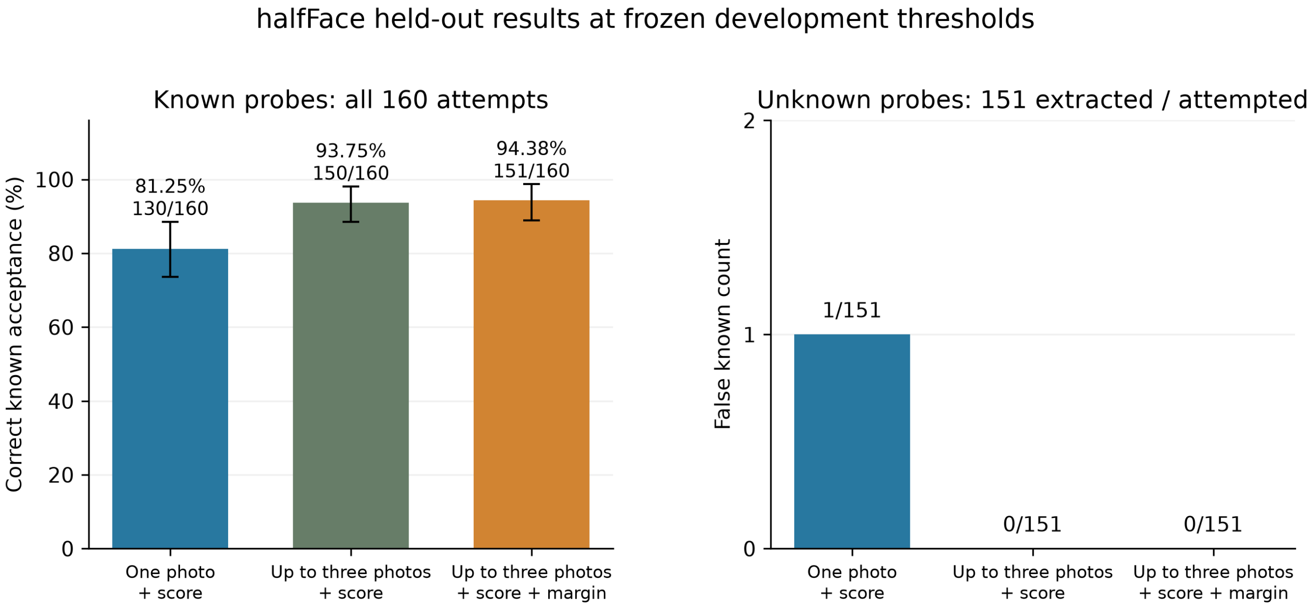
Method	Cosine threshold	Gap threshold	Quality gate
One photo / score	0.3887378871440888	0.0	0.0
Few photos / score	0.3978477716445923	0.0	0.0
Few photos / score+gap	0.3778589367866516	0.15	0.0

Table 4.3. Paired known-identity uncertainty, 2,000 resamples. Calibration/unknown-population uncertainty excluded.

Paired comparison	Change in points	95% interval in points
few photo score minus one photo score	12.500	6.92 to 18.75
few photo margin minus few photo score	0.625	-1.28 to 2.55
few photo margin minus one photo score	13.125	7.45 to 19.62

Ready-to-adapt text.

Up-to-three-photo score-only enrollment correctly accepted 150/160 masked known probes, compared with 130/160 for one-photo enrollment, an observed increase of 12.50 percentage points. The paired known-identity interval was 6.92 to 18.75 points. The primary few-photo score-and-margin method accepted 151/160 (94.38%), with 0/151 observed unknown false accepts. Its improvement over few-photo score-only decisions was one net probe, 0.625 points, with an interval from-1.28 to 2.55 points. This does not establish a clear additional margin benefit. Because each method uses its frozen development-selected operating point, these are calibrated-policy comparisons rather than a fixed-threshold isolation of the margin.


halfFace held-out results at frozen development thresholds

Known probes: all 160 attempts

Unknown probes: 151 extracted / attempted

Left: 95% descriptive known-identity bootstrap intervals; one capture failure stays in the denominator.
Right: zero observed false accepts is a sample result, not a population error bound. Primary method: few-photo + margin.

Figure 4.1. Frozen halfFace comparison. Correct known acceptance includes capture failures; intervals condition on chosen development settings. Unknown false-accept counts must accompany the percentages.

4.3.2 FINAL: outcome accounting and unknown rejection

Table 4.4. Primary method outcome counts. All 151 known acceptances have the correct identity.

Actual group	KNOWN	UNKNOWN	AMBIGUOUS	Quality refusal	Capture failure
Known 160	151	7	1	0	1
Unknown 151	0	151	0	0	0

The total is 311 probes: 151 KNOWN, 158 UNKNOWN, 1 AMBIGUOUS and 1 CAPTURE_FAILURE. Decision coverage is 309/311=99.36%; among 310 valid extractions it is 309/310=99.68%. Rank-1 retrieval is 159/160=99.38%, whereas correct known acceptance is 151/160=94.38%. There are no wrong-identity accepted known probes. Seven known people's probe images are rejected, one is withheld as ambiguous and one fails capture; do not hide these under a high covered-case accuracy.

The conditional and end-to-end unknown error rates are both 0/151 here because all unknown probes yield embeddings. This equality does not generally hold. An all-zero unknown-error bootstrap interval in saved output is not evidence of zero population risk. Repeated images from 27 unknown identities are not 151 independent unknown people. Report the sample and avoid a deployment guarantee.

4.3.3 ENGINEERING: reserved ICFD sample

Table 4.5. Five reserved public-sample identities; closed-set evidence only.

Enrollment	Gallery IDs / photos	Correct Rank-1 / attempts	Capture failures	Unknown probes
one photo	5 / 5	98/100 (98.00%)	2	0
up to three photos	5 / 15	98/100 (98.00%)	2	0

Both policies retrieve 98/100 correctly. The two misses are capture failures, and no unknown probes were included. Do not claim 98% open-set accuracy, full ICFD evaluation, or an enrollment-policy superiority result from this sample.

4.3.4 DEVELOPMENT: ArcFace and AdaFace checkpoint comparison

Table 4.6. Shared development probes: 154 known attempts(148 valid), 150 unknown attempts(149 valid).

Model	Original enrollment	Rule	Correct /154	Unknown FA /149	Valid coverage
arcface	One	Score	129/154	1/149	100.00%
arcface	One	Score+gap	139/154	1/149	98.99%
arcface	Up to 3	Score	138/154	1/149	100.00%
arcface	Up to 3	Score+gap	139/154	1/149	99.33%
adaface	One	Score	112/154	1/149	100.00%
adaface	One	Score+gap	120/154	1/149	95.96%
adaface	Up to 3	Score	122/154	1/149	100.00%
adaface	Up to 3	Score+gap	133/154	0/149	93.27%

Score-only few-photo acceptance improves for both released checkpoints. With separately calibrated score/margin settings, ArcFace one/few-photo policies tie at 139/154, while AdaFace changes 120 to 133/154. AdaFace few-photo gap decisions have lower valid coverage(93.27%) and 0/149 false unknown accepts, compared with ArcFace 99.33% coverage and 1/149. These differences are operating trade-offs, not proof that one loss function is superior. Both use identical SCRFD crops; pretraining and runtime engines differ.

Model	ICFD one-photo Rank-1	ICFD few-photo Rank-1	Scope
arcface	100/100	100/100	5 development IDs; no unknowns
adaface	99/100	99/100	5 development IDs; no unknowns

Table 4.7. Complete paired backbone-study effects; all descriptive development comparisons.

Comparison / rule	Change in points	Paired 95% interval
few_minus_one; arcface; up_to_three_photos; score_only	5.84	1.31 to 11.61
few_minus_one; arcface; up_to_three_photos; score_margin	0.00	-2.53 to 2.61
few_minus_one; adaface; up_to_three_photos; score_only	6.49	1.28 to 12.18
few_minus_one; adaface; up_to_three_photos; score_margin	8.44	1.37 to 15.79
adaface_minus_arcface; adaface; one_photo; score_only	-11.04	-18.18 to -4.67
adaface_minus_arcface; adaface; one_photo; score_margin	-12.34	-18.59 to -7.10
adaface_minus_arcface; adaface; up_to_three_photos; score_only	-10.39	-15.82 to -5.30
adaface_minus_arcface; adaface; up_to_three_photos; score_margin	-3.90	-8.75 to 0.65

4.3.5 DEVELOPMENT: score, margin and quality rules

Table 4.8. Main development rule ablation. The quality family selects a disabled gate.

Family	Correct /154	Unknown FA /149	Valid coverage	Selected quality
score_only	138	1	100.00%	0.0
score_margin	139	1	99.33%	0.0
score_margin_quality	139	1	99.33%	0.0

The score/margin and score/margin/quality families produce the same result because the selected quality gate is 0. Quality sensitivity can reduce coverage; it must not be presented as a proved improvement. The score gap here measures gallery ambiguity, not a new trained margin loss.

4.3.6 DEVELOPMENT: original-photo pooling

Table 4.9. Same candidates/probes. Fixed gap uses prior uniform-mean tau=.3778589367866516, gap=.15.

Pooling policy	Recalibrated correct /154	Recalibrated FA /149	Fixed-gap correct /154	Fixed-gap FA /149
one photo	129	1	131	0
uniform mean	138	1	139	1
heuristic weighted	138	1	138	1
detector weighted	139	1	138	1
best quality	129	1	131	0

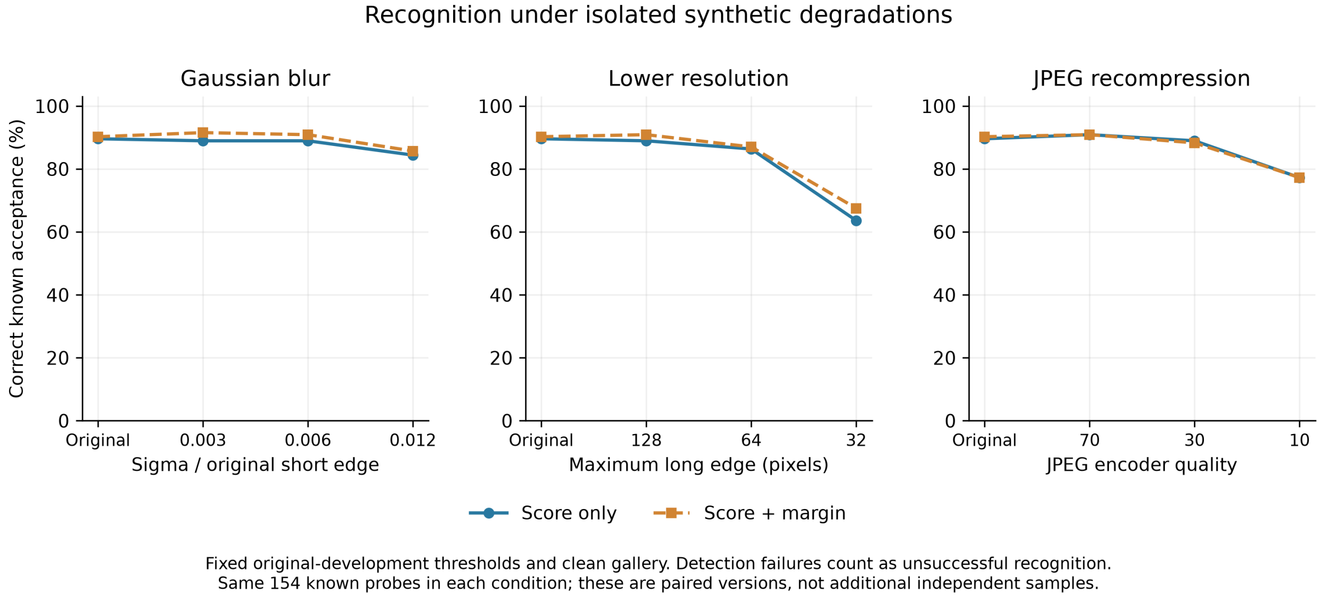
Detector weighting adds one acceptance over uniform mean under recalibrated score-only settings, but loses one under the fixed score/gap rule. The descriptive recalibrated interval is 0.00 to 2.04 points. There is no consistent weighting advantage. The one-photo fixed-rule 131/154 result differs from the backbone table's separately calibrated 139/154 because thresholds differ; these are not conflicting records. All five policies obtain 100/100 closed-set Rank-1 on the five-ID development ICFD sample, a ceiling check without unknowns.

4.3.7 DEVELOPMENT: robustness to probe transformations

Table 4.10. Original+9 single-factor conditions; same 304 attempts. FA denominators are valid unknowns.

Condition	Known valid	Unknown valid	Score correct	Gap correct	Score FA	Gap FA
original	148/154	149/150	138/154	139/154	1/149	1/149
blur_003	150/154	150/150	137/154	141/154	2/150	1/150
blur_006	152/154	150/150	137/154	140/154	1/150	1/150
blur_012	150/154	149/150	130/154	132/154	2/149	1/149
resolution_128	152/154	149/150	137/154	140/154	1/149	1/149
resolution_64	151/154	150/150	133/154	134/154	1/150	1/150
resolution_32	146/154	144/150	98/154	104/154	1/144	0/144
jpeg_70	149/154	149/150	140/154	140/154	1/149	1/149
jpeg_30	148/154	149/150	137/154	136/154	0/149	1/149
jpeg_10	130/154	145/150	119/154	119/154	0/145	0/145

The 32-pixel long-edge condition reduces gap-rule correct acceptance from 139 to 104/154. JPEG 10 gives 119/154, with 130/154 known and 145/150 unknown valid extractions. A lower false-accept count may accompany reduced utility. Mild transforms sometimes improve detector outcomes; do not invent monotonic performance loss or attribute all change to the recognizer. No transformed image is a new independent subject.


Recognition under isolated synthetic degradations

Gaussian blur

Lower resolution

JPEG recompression

Fixed original-development thresholds and clean gallery. Detection failures count as unsuccessful recognition.
Same 154 known probes in each condition; these are paired versions, not additional independent samples.

Figure 4.2. DEVELOPMENT: correct known acceptance under fixed-threshold stress tests. The source-image attempts remain paired; extraction outcomes can change.

4.3.8 DEVELOPMENT: synthetic enrollment, both threshold views

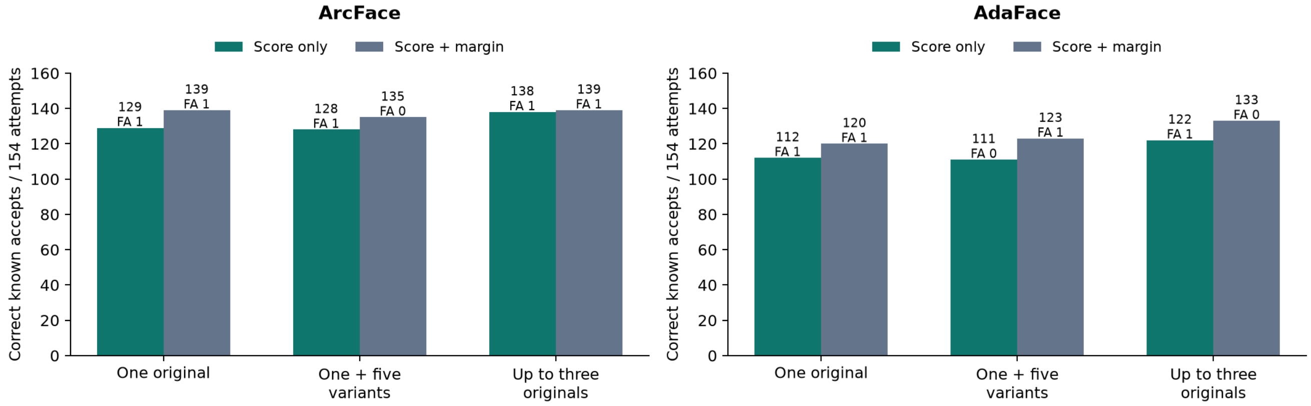
Table 4.11 a. Separately recalibrated on development. Same 154 known and 150 unknown attempts.

Model	Rule	Enrollment	Correct known	Unknown FA	Valid coverage
arcface	Score	One	129/154	1/149	100.00%
arcface	Score	One+5 variants	128/154	1/149	100.00%
arcface	Score	Up to 3 originals	138/154	1/149	100.00%
arcface	Gap	One	139/154	1/149	98.99%
arcface	Gap	One+5 variants	135/154	0/149	97.64%
arcface	Gap	Up to 3 originals	139/154	1/149	99.33%
adaface	Score	One	112/154	1/149	100.00%
adaface	Score	One+5 variants	111/154	0/149	100.00%
adaface	Score	Up to 3 originals	122/154	1/149	100.00%
adaface	Gap	One	120/154	1/149	95.96%
adaface	Gap	One+5 variants	123/154	1/149	95.29%
adaface	Gap	Up to 3 originals	133/154	0/149	93.27%

Table 4.11 b. Fixed previous one-photo settings. Same 154 known and 150 unknown attempts.

Model	Rule	Enrollment	Correct known	Unknown FA	Valid coverage
arcface	Score	One	129/154	1/149	100.00%
arcface	Score	One+5 variants	128/154	2/149	100.00%
arcface	Score	Up to 3 originals	138/154	1/149	100.00%
arcface	Gap	One	139/154	1/149	98.99%
arcface	Gap	One+5 variants	139/154	2/149	98.99%
arcface	Gap	Up to 3 originals	143/154	4/149	97.31%
adaface	Score	One	112/154	1/149	100.00%
adaface	Score	One+5 variants	111/154	2/149	100.00%
adaface	Score	Up to 3 originals	124/154	3/149	100.00%
adaface	Gap	One	120/154	1/149	95.96%
adaface	Gap	One+5 variants	123/154	1/149	95.29%
adaface	Gap	Up to 3 originals	139/154	5/149	92.93%

The recipe does not consistently replace independent original views. Under recalibrated score-only rules, ArcFace 129 to 128 and AdaFace 112 to 111/154 slightly worsen. Under gap rules, ArcFace 139 to 135 while AdaFace 120 to 123; the latter small paired interval touches zero. With fixed rules, additional originals can increase both known and false unknown acceptance, e.g.ArcFace 139 to 143 accompanied by 1 to 4/149 unknown errors. Report both views, not whichever favors a policy.



Descriptive development recalibration on the same probes; FA labels count false accepts among 149 extracted unknowns.

Figure 4.3. DEVELOPMENT: tested original-plus-five-variant recipe versus original-photo policies. Synthetic enrollment changes neither the probe set nor the number of independent participants.

Table 4.12. All 16 paired enrollment-extension effects.2,000 known-ID resamples; descriptive, fixed selected settings.

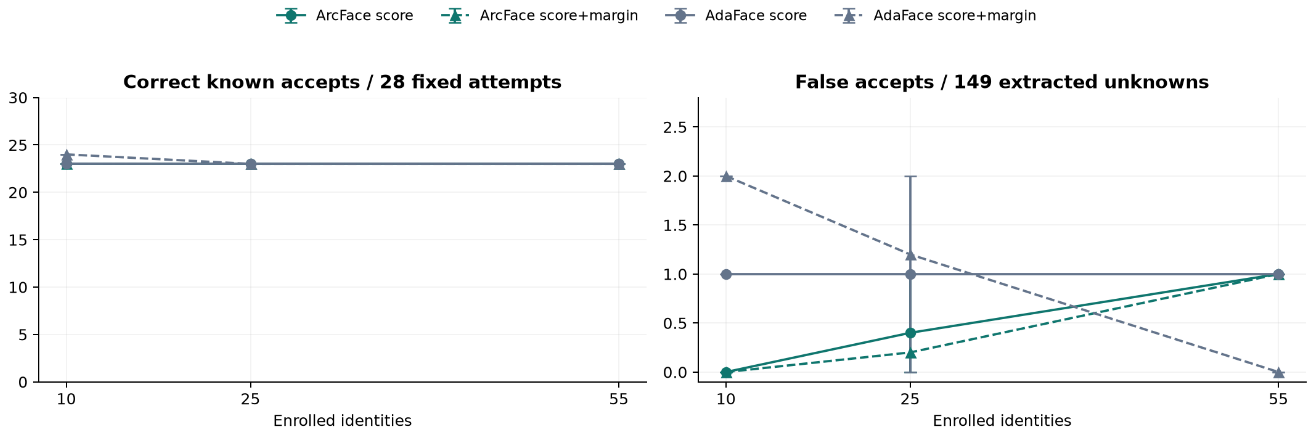
Model / rule / threshold view	Policy minus one original	Change points	Paired 95% interval
arcface / only / fixed	One+5 variants	-0.65	-2.01 to 0.00
arcface / only / fixed	Up to 3 originals	5.84	1.31 to 11.61
arcface / margin / fixed	One+5 variants	0.00	0.00 to 0.00
arcface / margin / fixed	Up to 3 originals	2.60	0.00 to 5.62
arcface / only / recal	One+5 variants	-0.65	-2.01 to 0.00
arcface / only / recal	Up to 3 originals	5.84	1.31 to 11.61
arcface / margin / recal	One+5 variants	-2.60	-5.19 to -0.63
arcface / margin / recal	Up to 3 originals	0.00	-2.53 to 2.61
adaface / only / fixed	One+5 variants	-0.65	-2.04 to 0.00
adaface / only / fixed	Up to 3 originals	7.79	2.04 to 13.84
adaface / margin / fixed	One+5 variants	1.95	0.00 to 4.40
adaface / margin / fixed	Up to 3 originals	12.34	5.88 to 19.62
adaface / only / recal	One+5 variants	-0.65	-2.04 to 0.00
adaface / only / recal	Up to 3 originals	6.49	1.28 to 12.18
adaface / margin / recal	One+5 variants	1.95	0.00 to 4.40
adaface / margin / recal	Up to 3 originals	8.44	1.37 to 15.79

4.3.9 DEVELOPMENT: fixed-probe gallery growth

Table 4.13. Complete 7-composition results per model; same 28 known/150 unknown attempts, 26/149 valid.

Model	Composition	Rank-1	Score correct	Score FA	Gap correct	Gap FA
arcface	g10	26/28	23/28	0/149	23/28	0/149
arcface	g25_seed42	26/28	23/28	1/149	23/28	0/149
arcface	g25_seed123	26/28	23/28	1/149	23/28	0/149
arcface	g25_seed2026	26/28	23/28	0/149	23/28	1/149
arcface	g25_seed7	26/28	23/28	0/149	23/28	0/149
arcface	g25_seed99	26/28	23/28	0/149	23/28	0/149
arcface	g55	26/28	23/28	1/149	23/28	1/149
adaface	g10	26/28	23/28	1/149	24/28	2/149
adaface	g25_seed42	26/28	23/28	1/149	23/28	0/149
adaface	g25_seed123	26/28	23/28	1/149	23/28	2/149
adaface	g25_seed2026	26/28	23/28	1/149	23/28	2/149
adaface	g25_seed7	26/28	23/28	1/149	23/28	0/149
adaface	g25_seed99	26/28	23/28	1/149	23/28	2/149
adaface	g55	26/28	23/28	1/149	23/28	0/149

Background composition changes competitors while the known core, probe images and operating thresholds stay fixed. Rank-1 remains 26/28. ArcFace acceptance stays 23/28, while AdaFace gap acceptance is 24/28 at 10 IDs and 23/28 otherwise. Unknown acceptance need not increase monotonically under a gap rule: an added competitor can produce ambiguity. Seed ranges describe these specific compositions, not confidence intervals or arbitrary-scale reliability.



Fixed probes/thresholds. At 25 identities: mean of five background compositions, with MIN/MAX whiskers (not confidence intervals).

Figure 4.4. DEVELOPMENT: gallery composition sensitivity with fixed probes and thresholds; 25-ID backgrounds vary, not the participant cohort.

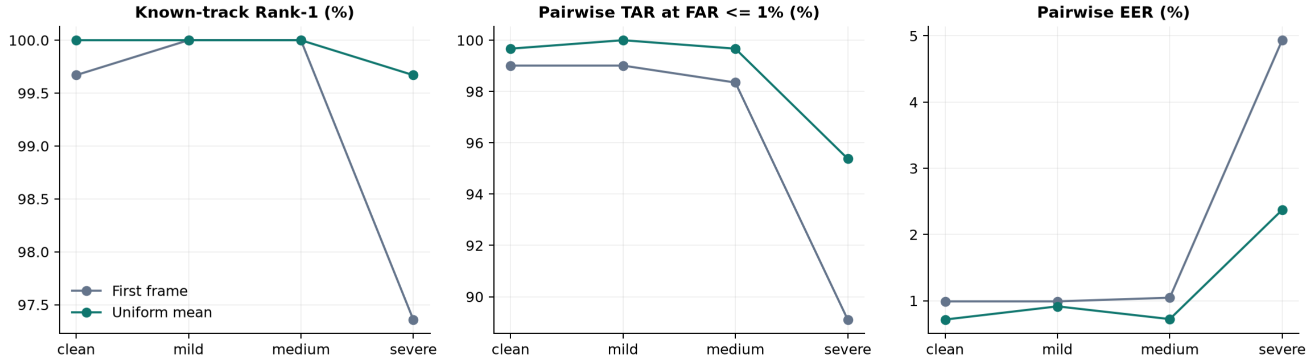
4.3.10 EXPLORATORY: temporal retrieval and threshold transfer

Table 4.14. EXPLORATORY: paired five-frame temporal study. Pairwise rates are descriptive evaluation ROC measures.

Tier	Policy	Rank-1	Pair TAR@1%	Pair EER	Known accepted	Unknown FA
clean	first	302/303	99.01%	0.99%	300/303	14/143
clean	mean	303/303	99.67%	0.71%	303/303	19/143
mild	first	303/303	99.01%	0.99%	298/303	9/143
mild	mean	303/303	100.00%	0.91%	303/303	10/143
medium	first	303/303	98.35%	1.04%	298/303	8/143
medium	mean	303/303	99.67%	0.72%	302/303	7/143
severe	first	295/303	89.11%	4.93%	223/303	1/143
severe	mean	302/303	95.38%	2.37%	282/303	4/143
mixed	first	302/303	95.38%	2.27%	291/303	13/143
mixed	mean	303/303	100.00%	0.66%	303/303	12/143

Ready-to-adapt text.

> Under severe synthetic corruption, mean pooling improves Rank-1 retrieval from 295/303 to 302/303, a 2.31-point difference with a paired interval of 0.96 to 3.75 points. This retrieval gain does not establish calibrated unknown rejection. For mixed tracks, the frozen mean rule accepts 303/303 known queries but also 12/143 unknown queries, an 8.39% observed FPIR, despite selecting thresholds to produce zero accepts among 66 unknown calibration tracks. Threshold transfer across the examined sequences is incomplete, and pairwise verification rates must not be interpreted as the same operating condition.



Paired evaluation tracks; supplied eye-aligned crops with synthetic corruption. Pairwise ROC summaries are descriptive.

Figure 4.5. EXPLORATORY: first-frame and uniform-mean performance under declared crop corruptions. This evaluates neither the detector nor a live tracker.

Table 4.15. All 9 declared mixed-track policies. No post-evaluation beta selection is claimed.

Mixed-track policy	Rank-1	Known accepted	Unknown FA
first	99.67%	291/303	13/143
random	99.34%	294/303	8/143
best_laplacian	98.68%	279/303	14/143
best_norm	99.67%	297/303	10/143
mean	100.00%	303/303	12/143
softmax_norm_0.5	100.00%	303/303	9/143
softmax_norm_1	100.00%	303/303	9/143
softmax_norm_2	99.67%	300/303	10/143
softmax_norm_5	99.67%	299/303	10/143

Table 4.16. Simulated one-frame identity substitution per known track(20%); identical intruders across methods.

Intruder control	Mean Rank-1	Best-norm Rank-1	Norm-softmax 1 Rank-1
none	303/303	302/303	303/303
matched	303/303	198/303	224/303
mild	303/303	171/303	205/303

A high-norm foreign frame can dominate selection or weighting. In the mild-intruder control, mean stays 303/303, best norm falls to 171/303 and norm-softmax 1 to 205/303. This is a synthetic sensitivity control, not a measured rate of tracking mistakes. Independent corruption noise and one portal limit generalization.

Table 4.17. Spearman correlations on 1,515 selected known evaluation frames per tier, 17 identities.

Tier	Mean norm	Laplacian/contrast	Norm/similarity r	Laplacian/similarity r
clean	21.10	0.2856	0.359	0.098
mild	21.46	0.2132	0.343	0.153
medium	21.58	0.1901	0.326	0.178
severe	19.51	0.3416	0.361	0.306

The Laplacian/contrast mean rises from.1901 at medium to.3416 at severe corruption even as severe noise degrades recognition; raw norm drops 21.58 to 19.51. Within-tier Laplacian correlations remain positive. Report noise sensitivity and limited usefulness, not a universal statement that the proxy is useless. Nominal frame-level p-values are not independent-sample evidence.

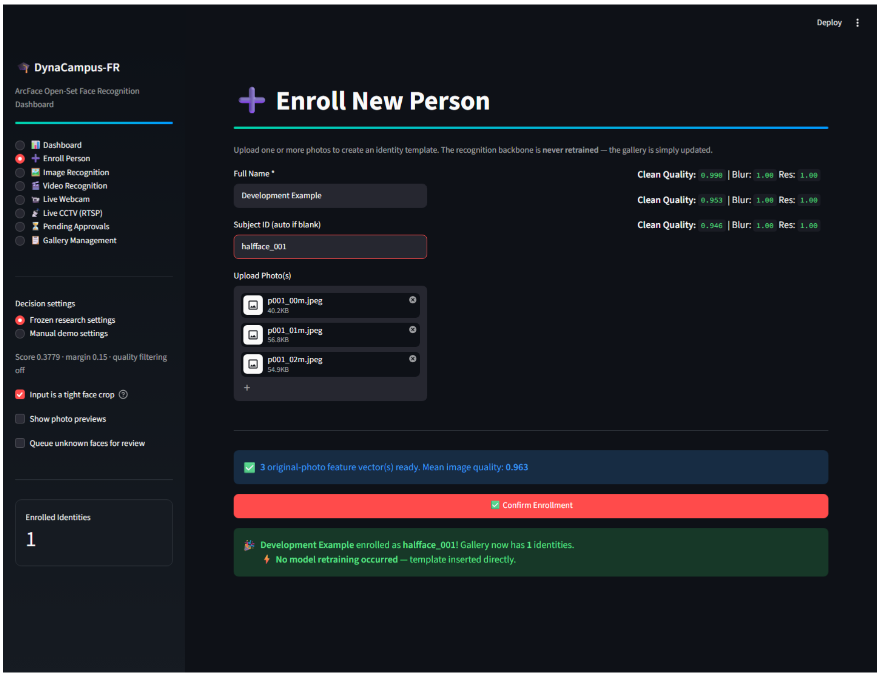
Table 4.18. Separate calibration for fixed/floored adaptive rules; evaluation FPIR is not matched.

Condition	Rule	Known accepted	Unknown FA	Base threshold
severe	fixed	282/303	4/143	0.237205
severe	aosm	282/303	3/143	0.281290
mixed	fixed	303/303	12/143	0.214824
mixed	aosm	303/303	13/143	0.229286

The adaptive rule changes severe false acceptance 4 to 3/143 while known acceptance stays 282/303; on mixed queries it changes 12 to 13/143 with 303 known accepted. It is not a consistent rejection improvement and must not be retroactively substituted into the main final method. Complete policy/condition summaries and paired intervals are included in the appendix.

4.3.11 ENGINEERING: persistent enrollment and dashboard evidence

The original dashboard's saved development demonstration inserts 55 identities without changing the recognition checkpoint, verifies persistence and removal, and checks original upload bytes against the research path. A separate defense walkthrough starts with 54 cached real development templates and freshly processes target enrollment photos and every selected probe. It demonstrates known, unknown, ambiguous and capture-failure cases across ten steps, then removes the demonstration target. These are disclosed examples, not extra accuracy samples.



Decision settings
● Frozen research settings
○ Manual demo settings

Score 0.3779 · margin 0.15 · quality filtering off

☑ Input is a tight face crop
☐ Show photo previews
☐ Queue unknown faces for review

Enrolled Identities: 1

Upload one or more photos to create an identity template. The recognition backbone is **never retrained** — the gallery is simply updated.

Full Name *: Development Example

Subject ID (auto if blank): halfface_001

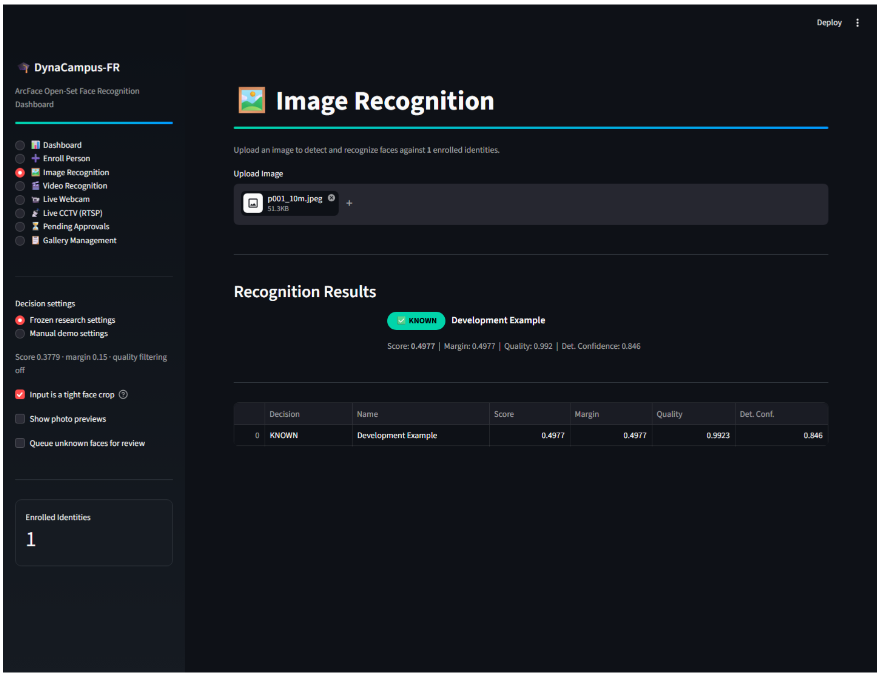
Upload Photo(s):
p001_00m.jpeg 40.2KB
p001_01m.jpeg 56.8KB
p001_02m.jpeg 54.9KB

Clean Quality: `0.990` | Blur: `1.00` Res: `1.00`
Clean Quality: `0.953` | Blur: `1.00` Res: `1.00`
Clean Quality: `0.946` | Blur: `1.00` Res: `1.00`

✓ 3 original-photo feature vector(s) ready. Mean image quality: **0.963**

Development Example enrolled as **halfface_001**! Gallery now has **1** identities.
No model retraining occurred — template inserted directly.

Figure 4.6. Historical actual application enrollment screenshot with photo previews off. It documents interface behavior, not new performance evaluation.


Deploy
DynaCampus-FR
ArcFace Open-Set Face Recognition Dashboard
○ Dashboard
○ Enroll Person
● Image Recognition
○ Video Recognition
○ Live Webcam
○ Live CCTV (RTSP)
○ Pending Approvals
○ Gallery Management
Image Recognition
Upload Image
Recognition Results

Upload an image to detect and recognize faces against **1** enrolled identities.

p001_10m.jpeg 51.3KB

Decision settings

● Frozen research settings
○ Manual demo settings

Score 0.3779 · margin 0.15 · quality filtering off

☑ Input is a tight face crop
☐ Show photo previews
☐ Queue unknown faces for review

Enrolled Identities 1

✓ KNOWN **Development Example**

Score: **0.4977** | Margin: 0.4977 | Quality: 0.992 | Det. Confidence: 0.846

	Decision	Name	Score	Margin	Quality	Det. Conf.
0	KNOWN	Development Example	0.4977	0.4977	0.9923	0.846

Figure 4.7. Historical actual still-image KNOWN example. A selected example is not an independent benchmark or a live-camera claim.

Table 4.19. Gallery lifecycle verification; selected development examples.

Lifecycle step	Expected evidence	Established scope
Before target enrollment	Target absent; gallery has 54 background IDs.	Isolated demonstration gallery.
Enroll	Fresh original images form target template; 55 IDs.	No backbone training/hash change.
Recognize	Selected known/unknown/ambiguous/failure outcomes.	Fresh features reproduce saved real development features.
Reload	Same ID set and exact stored templates after reload.	File persistence; no SQL event database.
Remove	Target gone; 54 backgrounds remain.	Removal correctness, not a legal retention certification.

Current saved defense receipt marks fresh browser capture and live screen recording false. The backup video assembles existing evidence; it is not a freshly recorded camera session. Webcam/RTSP pages and SORT-like code exist, but a camera benchmark, tracking identity metrics and multi-camera system remain unvalidated. Do not substitute rendering a control for testing its operational behavior.

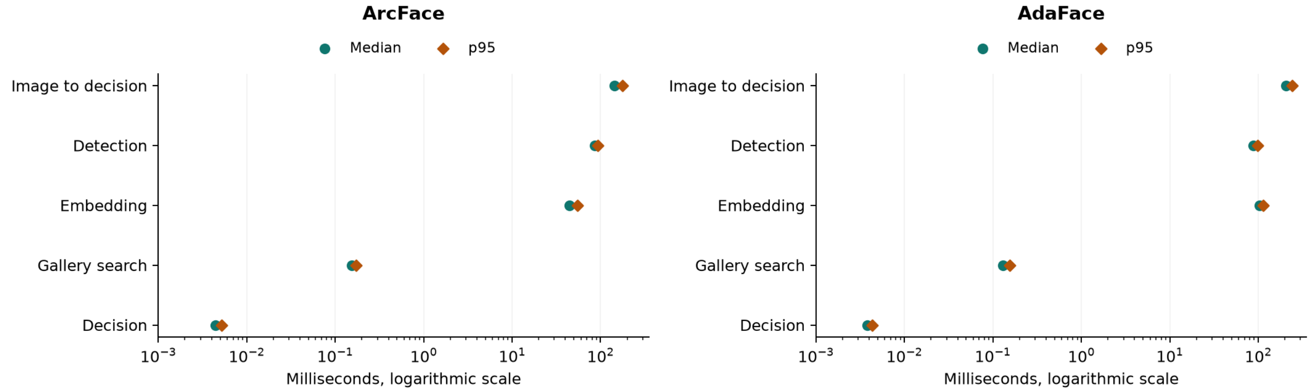
4.3.12 ENGINEERING: measured CPU latency

Table 4.20. Direct wall-time enrollment/recognition and selected components; 55-ID baseline gallery.

Model	Measured boundary / stage	Calls	Median ms	p95 ms
arcface	enrollment / enrollment_end_to_end	30	136.9683	173.1506
arcface	enrollment / persistent_insertion	30	1.8721	2.7116
arcface	recognition / recognition_end_to_end	60	143.8087	175.5687
arcface	recognition / detection	60	86.3946	92.5805
arcface	recognition / embedding	60	44.1930	55.0558
arcface	recognition / gallery_search	60	0.1542	0.1711
arcface	recognition / decision	60	0.0044	0.0052
adaface	enrollment / enrollment_end_to_end	30	199.8750	221.3193
adaface	enrollment / persistent_insertion	30	1.5802	2.1713
adaface	recognition / recognition_end_to_end	60	207.0253	241.8298
adaface	recognition / detection	60	88.2088	97.9901
adaface	recognition / embedding	60	103.7119	113.0020
adaface	recognition / gallery_search	60	0.1299	0.1541
adaface	recognition / decision	60	0.0038	0.0043

Table 4.21.300 actual search/decision calls per model/size; matrix assembly/sorting included.

Model	Gallery IDs	Actual search median/p95 ms	Decision median/p95 ms
arcface	10	0.0296 / 0.0438	0.0011 / 0.0015
arcface	25	0.0408 / 0.0421	0.0011 / 0.0013
arcface	55	0.0637 / 0.0708	0.0012 / 0.0014
adaface	10	0.0293 / 0.0402	0.0012 / 0.0015
adaface	25	0.0408 / 0.0438	0.0012 / 0.0013
adaface	55	0.0635 / 0.0695	0.0012 / 0.0013



60 warm real-image recognition calls per model; CPU inference threads=2, 55-identity gallery. No browser, camera capture or network timing.

Figure 4.8. ENGINEERING: CPU timing stages. Direct total wall times are measured separately from component medians; model engines differ.

Ready-to-adapt text.

Image processing dominates search in the tested small galleries. ArcFace recognition has a median 143.81 ms and p95 175.57 ms, whereas its 55-ID search microbenchmark median is approximately.064 ms. AdaFace recognition is 207.03 ms median and 241.83 ms p95 in its PyTorch CPU implementation. The gap cannot be attributed solely to a loss function because engines and checkpoints differ. Measurements use repeated fixed development inputs with warm CPU/filesystem conditions and exclude model loading, browser transport, camera capture and RTSP/network time. No measured camera FPS follows from these timings.

4.3.13 Integrated discussion and contribution boundary

Ready-to-adapt text.

The strongest final evidence concerns additional original enrollment views: few-photo score-only enrollment improves correct known acceptance by 20 probes over the one-photo baseline at separately frozen calibrated settings. The margin rule adds only one net probe, with an interval including zero. The development studies show that these effects depend on the recognizer, operating rule and coverage. The declared synthetic-enrollment recipe does not consistently replace independent original views, and simple quality weights do not reliably outperform the mean. The temporal supplement separates retrieval benefit from unknown rejection: averaging can improve severe synthetic retrieval while calibration fails to transfer at its intended error target. Together these results support transparent paired baselines, complete outcome reporting and representative future calibration rather than universal superiority claims.

Explain threats to validity explicitly: public-data capture conditions; possible unobserved pretraining overlap and near duplicates; small unknown identity counts; masked-known/mixed-condition unknowns; development reuse; conditional bootstrap intervals; composition repeats; one portal and correlated frames; independent synthetic noise; camera/sequence confounding; checkpoint/engine confounding; warm timing; absence of campus, fairness, spoofing and live-tracker evaluation. These limits do not invalidate the measured results, but restrict the claims they support.

4.4 Summary

Summarize what was learned rather than repeat every table.

Ready-to-adapt text.

The frozen final comparison supports a measured few-photo enrollment benefit, while the incremental margin effect is small and uncertain. Development ablations expose checkpoint, pooling, augmentation and gallery-composition interactions. The exploratory temporal study demonstrates averaging benefits under severe synthetic corruption alongside a failure of unknown-threshold transfer. Software and numerical audits support implementation consistency, and the engineering measurements establish local image/gallery costs without implying representative real-time campus operation.

Chapter 5. Standards and Design Constraints

Ready-to-adapt text.

This chapter examines the technical conventions, resource choices and social constraints relevant to the research prototype. It also presents an explicit alternative-cost analysis and proposed mappings to complex problem-solving and engineering-activity categories, with the limits of the evidence stated.

5.1 Compliance with the Standards

The template asks for relevant standards, alternatives, pros/cons and selection rationale. Distinguish using an interoperable format/protocol from being certified against a standard. There is no established ISO certification, accredited security audit or biometric-standard conformance test for this prototype. Do not invent one.

5.1.1 Software Standards

Discuss the actual formats, interfaces and engineering conventions used; add pros/cons and rationale. Model/data licenses are obligations to respect, not performance certifications.

Choice and alternative	Advantages	Limit / rationale
ONNX inference vs native-training framework	Portable inference model and explicit CPU provider for existing ArcFace/SCRFD assets.	Chosen for available checkpoints; AdaFace uses PyTorch, so engine differences confound comparisons.
JSON/NPZ vs SQL/vector service	Readable metadata and compact numerical storage; simple local auditing.	Selected for small isolated galleries; no multiuser transaction, access-control or durable campus event-history guarantee.
Config files + frozen hashes vs ad hoc UI settings	Preserves exact experimental choices and input provenance.	Manual demo settings are permitted in UI but cannot change reported frozen methods.
Unit tests + separate result audits vs screenshots alone	Checks failure handling, mathematical operations and stored numerical consistency.	Tests do not certify biometric generalization or security.
Documented source/model terms vs treating all downloads as unrestricted	Retains provenance and attribution.	Source-code license, model license and dataset permissions differ; commercial deployment is not established.

Suggested prose: "The project uses standardized file formats and explicit software interfaces to make the prototype inspectable. Reproducible configurations, model/source hashes and separate verification scripts support traceability. These practices are software quality measures, not a claim of compliance certification or institutional authorization."

5.1.2 Hardware Standards

Explain why CPU inference and existing hardware were selected. The report does not establish a formal minimum-memory or minimum-processor requirement.

Alternative	Pros	Cons / selected scope
Existing CPU workstation	No accelerator required for reported runs; measured local cost boundaries.	Approximately 144 ms ArcFace image-recognition median on tested setup is not a universal speed guarantee.
GPU workstation	Potential parallel/throughput improvements.	Not evaluated; extra acquisition/power/driver requirements; do not claim measured gains.
Edge device / mobile platform	Potential local processing and proximity to camera.	No quantization, edge-model or embedded timing benchmark completed.

Alternative	Pros	Cons / selected scope
Single-stream prototype vs multi-camera installation	Simplifies local development and demonstration.	No multi-camera synchronization, sustained load or capacity result.

5.1.3 Communication Standards

This section is present in the original template and must not be omitted. Discuss what is implemented and what is only a prototype control.

- Local browser/server interaction uses the Streamlit web application. The provided launcher binds to 127.0.0.1. This is a local development setup, not an authenticated internet service.

- OpenCV camera/video paths and RTSP URL controls exist. The tested evidence does not establish live RTSP transport reliability, camera compatibility or secure credential handling.

- Alternative 1: local file/image uploads offer deterministic reproducible inputs but do not measure network latency. Alternative 2: a live stream offers continuous frames but adds capture, transport, disconnect and authentication issues.

- A future remote service would need protected transport, authentication, credential handling and network failure policies. These are future requirements, not controls demonstrated by this thesis.

- The main reported image timings exclude communication/UI/capture; do not combine them with an assumed camera frame rate.

5.2 Design Constraints

Keep every category from the supplied template, including Political Constraint. A category with limited observed impact can state that limitation rather than being filled with a fabricated incident.

5.2.1 Economic Constraint

Discuss limited compute, reuse of equipment, public-sample access and the cost of obtaining representative data. CPU experiments and file galleries reduce prototype infrastructure needs but do not make deployment free. Separate capital spending, operating costs and in-kind resources; no actual expenditure total is established by code artifacts.

5.2.2 Environmental Constraint

Distinguish the image-capture environment from ecological impact. Masks, lighting, pose, resolution, compression and frame quality affect recognition. Controlled stress tests approximate selected changes but do not recreate campus camera height, crowds, weather or every codec. Environmental sustainability is treated separately in 5.2.7.

5.2.3 Ethical Constraint

Explain the sensitivity of images and embeddings, potential exclusion from false rejection, false identification, human review and function creep. Local storage and hidden previews reduce casual exposure but do not prove encryption, retention governance or consent compliance.

Ready-to-adapt text.

> The study uses public research inputs with their source terms and separates identifying media from aggregate paper figures. Recognition outputs are treated as uncertain research decisions rather than automated access permissions. The prototype does not include anti-spoofing, institutional authentication or a verified retention/deletion policy. Before collecting a local campus cohort or using the system operationally, the institution would need appropriate participant communication, approvals, access restrictions and a review process. South Asian dataset coverage does not by itself demonstrate demographic fairness.

5.2.4 Health and Safety Constraint

A false identity decision can lead to an inappropriate intervention, exclusion or misplaced security confidence. The research prototype does not actuate a gate or make disciplinary decisions. Describe capture failures and refusals as routes for human review. For future physical cameras, safe mounting, power/cabling and accessibility are installation requirements, not tested outcomes of the current software study.

5.2.5 Social Constraint

Discuss acceptance of biometric monitoring, unequal capture conditions, mistaken identity, consent choices and accessibility. Do not claim improved campus trust or student acceptance without a user study. Proposed mitigation: clearly communicated scope, operator review, alternatives to face recognition and limited data retention; label those as future governance design.

5.2.6 Political Constraint

Use the exact template category. No political-event study or policy-impact measurement was performed. Relevant design concerns are how institutional authority, surveillance policy, procurement and restrictions on data movement could affect future deployment. Explain that an academic prototype does not authorize monitoring political activity or inferring affiliation. Do not invent legal advice, political incidents or current-law compliance; describe the need for institutional policy review before operation.

5.2.7 Sustainability

State defensible design considerations: reuse hardware, frozen inference rather than repeated large-model training, and avoid redundant copies of sensitive media. The system still consumes electricity and storage. No power meter, carbon accounting or lifecycle assessment was used; therefore do not report quantified energy/emissions savings. Future comparison should measure whole-pipeline power, equipment lifecycle and load rather than infer sustainability from parameter count.

5.3 Cost Analysis

The template explicitly requires a budget, an alternative budget with rationale, and a revenue model. The following worked comparison is a transparent hypothetical planning example, NOT actual expenditure, market quotations or permission to commercialize. Replace its allowances with confirmed invoices/quotes in the final submission; retain the structure and rationale.

Proposed budget comparison for Section 5.3. All numerical amounts are illustrative assumptions, not researched purchase prices.

Planning item	A: reuse existing PC, BDT	B: dedicated PC, BDT	Assumption
Computer acquisition	0	50000	A assumes an already owned computer; B assumes a planning allowance only.
Camera allowance	2500	2500	Future demo/deployment purchase, not a camera used in reported timing.
Storage/backup allowance	1500	1500	Hypothetical extra storage.
Power/network allowance	600	600	Illustrative project-period allowance, not measured bills.
Printing/binding allowance	1400	1400	Illustrative submission cost.
Subtotal	6000	56000	Arithmetic from the above assumptions.
10% contingency	600	5600	Planning reserve.
Illustrative total	6600	61600	Not a statement of actual cash spent.

AlternativeA fits a research prototype when suitable equipment already exists; it has lower incremental capital cost but depends on shared hardware availability. AlternativeB improves equipment availability but has greater upfront cost and still needs governance, support and operational evaluation. Software development time, supervision, institutional infrastructure and dataset licensing are additional real resources even when no invoice is paid. State unavailable actual amounts as "not recorded", not 0.

Project cost = new equipment + operating costs + data/permissions + labor + submission + contingency

Annual net cash flow (future scenario) = institutions * annual service fee - annual operating/support cost

Revenue model. The present project is an academic research prototype with no demonstrated sales, customers or revenue. A future institution-supported/internal service could recover development and maintenance costs, or a licensed service could charge installation/support fees only after suitable rights, evaluation and governance are secured. Do not propose selling restricted research datasets/checkpoints. For a purely illustrative break-even calculation, 3 institutions payingBDT12,000/year givesBDT36,000 gross; if support/operation costsBDT30,000/year, the remainder isBDT6,000/year before tax, development recovery and other costs. This is arithmetic under assumed values, not a forecast or evidence of commercial feasibility. An academically acceptable conclusion is that commercialization was not evaluated.

5.4 Complex Engineering Problem

Provide actual project mappings instead of copying check marks from the blank template. The template uses P1-P7/A1-A5 terminology. The category names below follow that template and BAETE's corresponding legacy mapping reference[R11]. BAETE's newer published framework uses WP/EA/WK notation[R12]; do not claim an accreditation judgment. Keep the supplied template labels unless the department instructs otherwise.

5.4.1 Complex Problem Solving

The following is a proposed mapping with evidence and limits. P1 is the central justification. Other categories are not automatically satisfied merely because machine learning is involved.

Proposed Table 5.1 summary row: P1=yes; P2=yes; P3=yes; P4=qualified; P5=limited; P6=potential/qualified; P7=yes. Explain each in a subsection, as the template explicitly requests. "Yes" means a defensible academic rationale, not official assessor approval.

Table 5.1 expanded. Complex problem-solving mapping with evidence-based qualifications.

Category	Proposed mapping	Project-specific rationale / limit
P1 Depth of knowledge	Supported	Embedding geometry, uncertainty, calibration, software interfaces and experimental leakage controls require more than running a classifier. See K3/K4/K5/K6/K8 mapping below.
P2 Conflicting requirements	Supported	Correct known acceptance competes with unknown rejection, coverage and processing cost; show the measured trade-offs rather than asserting all improve.
P3 Depth of analysis	Supported	Choosing templates and thresholds requires paired experiments and careful denominators; high Rank-1 alone cannot solve open-set operation.
P4 Familiarity of issues	Qualified	Cross-sequence calibration failure, duplicate-ID labels and high-quality intruders are nontrivial for this project; face recognition itself is an established field.
P5 Applicable codes	Limited	An application-specific operating point and protocol are not fully prescribed by a generic software code. This does not mean the project is outside legal/ethical obligations or has created a standard.
P6 Stakeholder involvement	Potential/qualified	Students, operators, administrators and researchers could have conflicting needs, but no deployment consultation/user study is documented. Distinguish stakeholder analysis from measured involvement.
P7 Interdependence	Supported	Detection, alignment, embeddings, gallery composition, rejection and interface/persistence interact; upstream failure affects end-to-end utility.

P1 knowledge-profile mapping and rationale

The original template explicitly asks for an additional knowledge-profile mapping for P1. The table below covers K1-K8 and identifies which dimensions support P1 directly. It is not necessary to assert equal depth for every dimension.

Additional P1 mapping required by the supplied template. Category labels summarized from[R11]; applications are project analysis.

Profile	Connection to this project	Strength / P1 role
K1 Natural-science foundations	Image formation, illumination and blur provide context. No optical-physics investigation performed.	Supporting,limited.
K2 Mathematics / statistics	Vector normalization, cosine geometry, empirical operating points and identity-cluster bootstrap.	Strong analytical support.
K3 Engineering fundamentals	Modular pipelines, interfaces, validation and propagation of capture failures.	Direct P1 support.
K4 Specialist knowledge	Pretrained face representations, open-set identification and quality/temporal behavior.	Direct P1 support.
K5 Design knowledge	Trade-offs in templates, decision rules, storage and experimental controls.	Direct P1 support.
K6 Engineering practice	Real ONNX/PyTorch inference, persistent galleries, CPU measurement and regression checks.	Direct P1 support.
K7 Societal/professional context	Biometric risks, resource constraints and limits on surveillance claims.	Documented design reasoning; no certified compliance assessment.
K8 Research literature	Select and attribute established methods; formulate comparisons and interpret negative results.	Direct P1 support.

Ready-to-adapt text.

P1 is supported by the combination of specialist recognition knowledge and first-principles analysis of normalized descriptors, gallery matching and decision operating points. K3 supports the handling of interacting pipeline stages and failure propagation; K4 supports the recognition and open-set concepts; K5 supports the design of alternative templates and rejection rules; K6 supports real-model integration and reproducible measurement; and K8 supports the selection and critical interpretation of prior methods. These connections are evidenced by the paired experiments and verification procedures rather than by an unsupported claim of a new recognition algorithm.

Rationale subsection instructions. Under 5.4.1, use short labeled paragraphs/subsections for P1 throughP7. For each supported category, connect one design problem, one concrete artifact/result and one limitation. For qualified categories, explicitly say what was analyzed and what was not demonstrated. Do not state that all stakeholders were interviewed, that deployment caused significant consequences, or that the work is novel simply to secure every tick.

5.4.2 Engineering Activities

Use Table 5.2 with the template's A1-A5 labels and add a rationale paragraph/subsection for each category. Creative application is different from inventing a new recognition method.

Table 5.2 expanded. Engineering-activity mapping with rationale and non-deployment limits.

Activity	Proposed mapping	Project-specific rationale / limit
A1 Resource range	Supported	Integrates people's work, public datasets, source/checkpoint assets, CPU tools, storage and experimental documentation; no fabricated procurement claims.
A2 Interaction	Supported	Resolve image/protocol/backend/interface interactions and acceptance-versus-rejection trade-offs; shared crops and complete failure counts expose them.
A3 Innovation	Qualified,experimental	The controlled comparisons, correction of protocol issues and evidence integration are creative engineering work. Existing averaging/losses/dynamic enrollment are not claimed as inventions.
A4 Consequences	Potential,analyzed	Misidentification and surveillance could affect people and resources; actual deployed societal effects were not measured. Explain risk/refusal reasoning without inventing impact.
A5 Familiarity	Supported in team learning	Integration and debugging extend the project team's experience through principles-based analysis. Do not call established face recognition a wholly unfamiliar research field.

Ready-to-adapt text.

A1 and A2 are illustrated by combining public data, pretrained inference engines, a persistent gallery and an operator interface while retaining consistent preprocessing and comparison roles. A3 is supported as creative experimental integration and analysis, not as algorithmic novelty. A4 is addressed through analysis of possible biometric harms and clear refusal states; no deployed societal benefit is claimed. A5 reflects the team's extension of prior experience through documented design alternatives, troubleshooting and reproducible evaluation.

5.5 Summary

Close the chapter by distinguishing design reasoning from evidence of operational compliance.

Ready-to-adapt text.

The project selects local CPU inference, inspectable storage and explicit experiment controls within limited resources. Its constraints include public-data representativeness, biometric sensitivity, uncertain deployment behavior and unmeasured operational costs. The proposed budget alternatives and engineering mappings explain the design choices without claiming actual revenue, certification or campus impact. Responsible operational use requires further evaluation, institutional governance and suitable data/model permissions.

Chapter 6. Conclusion

Ready-to-adapt text.

This chapter summarizes the supported contributions, identifies the limitations of the experiments and prototype, and outlines future work. The conclusions distinguish measured findings from deployment aspirations.

6.1 Summary

Answer the objectives using evidence; do not introduce a new experiment or reuse old training accuracy as a current headline.

Ready-to-adapt text.

This project investigated few-photo enrollment and open-set identification using frozen pretrained recognition models. The main identity-disjoint halfFace comparison found that up-to-three-photo score-only enrollment correctly accepted 150/160 masked known probes compared with 130/160 for one-photo enrollment. The primary score-and-margin policy accepted 151/160, with 0/151 observed unknown false accepts, but the additional one-probe effect over few-photo score-only decisions was uncertain. Development studies showed interactions among recognizer, template policy and decision rule, with no consistent benefit from the tested synthetic-enrollment profile or heuristic quality weighting. In the exploratory temporal supplement, averaging improved severe synthetic retrieval while unknown-threshold transfer remained inadequate. The persistent gallery and measured CPU costs demonstrate a working research implementation. The contribution is the reproducible experimental evidence, explicit outcome accounting and integration of existing components, rather than a new recognition architecture or validated campus security system.

6.2 Limitation

Use explicit numbered limitations. The following covers the major threats required to interpret every result.

- **Data:** Public image conditions differ from campus CCTV; halfFace known probes are masked whereas unknown conditions are mixed. Local exact-hash/identity checks do not exclude near duplicates or external pretraining overlap.
- **Unknown population:** The final study includes 27 unknown IDs and 151 images, not 151 independent people; observed zero errors does not establish a population zero-error bound.
- **Selection and uncertainty:** Later studies reuse development calibration/assessment. Paired intervals condition on selected thresholds and omit several sources of uncertainty; exploratory comparisons are not a systematic significance screen.
- **Checkpoint comparison:** Pretraining, architecture and runtime engines differ; no isolated loss-function or detector-superiority conclusion follows.
- **Manipulations:** Synthetic blur/noise/compression and enrollment variants are limited recipes, not independent collected views or complete real-world camera simulation.
- **Temporal:** One portal, shared identity populations, prior pilot exposure and camera/sequence confounding limit the ChokePoint experiment. It bypasses detection and actual tracking.
- **Engineering:** Warm fixed-input timings exclude loading, camera, UI andnetwork; small galleries do not establish scale or sustained stream capacity.
- **Deployment:** No collected campus cohort, demographic fairness evaluation, liveness, validated domain adaptation, multi-camera operation, attendance integration or production security certification is established.
- **Transformer:** FaceLiVT assets and drafts exist, but no verified completed comparison is supported by the reviewed state.

6.3 Future Work

Tie each proposal to a measured limitation and state the evidence required. Do not list completed ArcFace/AdaFace inference as wholly future work.

- Complete a pinned, tested FaceLiVT inference/comparison path and evaluate it first on development data; any fresh final claim requires genuinely unexamined, appropriately audited evaluation data.
- Collect representative institution-approved enrollment/probe/camera data with explicit roles, unknown coverage and justified permissions; evaluate capture conditions and demographic performance rather than infer fairness.
- Measure continuous camera throughput, end-to-end latency, memory, utilization, disconnect recovery and tracker identity metrics; then investigate multi-camera synchronization.
- Validate calibration transfer with sufficient independent unknown identities, coverage targets and new sites/cameras. Do not retune the consumed final set and call it a new test.
- Investigate real paired-domain learning and trained quality estimates using leakage-controlled training/validation/evaluation. Random-vector adapter training is not a substitute.
- Add and evaluate anti-spoofing/human review and institutional authentication, protected event storage, retention/deletion controls and attendance integration.
- Measure power/lifecycle costs and deployment economics before claiming energy efficiency or commercial feasibility.

Section completion check: Every future item is visibly future; final conclusion matches the abstract and title.

Embedded supplementary result tables

These tables provide the remaining reporting-level numeric detail without requiring external spreadsheets. They summarize saved experiments, not new data. Raw per-image lists and thousands of calibration candidates are audit artifacts rather than necessary thesis-body tables; they are not reproduced as fictitious additional samples. Keep the final headline in Chapter 4 and place these detailed tables in an appendix if needed.

A1. Complete temporal policy outcomes

EXPLORATORY. Each row uses 303 known and 143 unknown evaluation tracks. R1=correct Rank-1 known tracks. F=calibration-frozen fixed rule; A=calibration-frozen floored adaptive rule. Known columns require both correct identity and acceptance. No column is an independently matched evaluation FPIR condition. Codes: bn=best norm; bl=best Laplacian; sm=norm-softmax with shown beta.

Tier	Intruder	Policy	R1 /303	F known /303	F FA /143	A known /303	A FA /143
clean	none	first	302	300	14	300	10
clean	none	random	303	302	15	302	12
clean	none	bn	303	302	15	302	13
clean	none	bl	302	300	10	300	9
clean	none	mean	303	303	19	303	13
clean	none	sm.5	303	303	20	303	12
clean	none	sm1	303	303	18	303	13
clean	none	sm2	303	303	14	302	13
clean	none	sm5	303	302	14	302	13
mild	none	first	303	298	9	299	7
mild	none	random	303	302	8	302	5
mild	none	bn	303	303	12	303	12
mild	none	bl	303	301	8	298	5
mild	none	mean	303	303	10	303	7
mild	none	sm.5	303	303	11	302	7
mild	none	sm1	303	303	12	302	8
mild	none	sm2	303	302	12	302	10
mild	none	sm5	303	303	13	302	13
medium	none	first	303	298	8	298	5
medium	none	random	302	299	4	300	7
medium	none	bn	303	299	4	299	5
medium	none	bl	302	300	6	301	3
medium	none	mean	303	302	7	303	6

Tier	Intruder	Policy	R1 /303	F known /303	F FA /143	A known /303	A FA /143
medium	none	sm.5	303	301	6	301	6
medium	none	sm1	303	300	5	300	5
medium	none	sm2	303	300	4	300	5
medium	none	sm5	303	300	4	300	4
severe	none	first	295	223	1	190	0
severe	none	random	293	222	0	194	0
severe	none	bn	294	252	4	251	5
severe	none	bl	292	258	8	246	5
severe	none	mean	302	282	4	282	3
severe	none	sm.5	302	286	7	283	7
severe	none	sm1	302	283	8	282	7
severe	none	sm2	298	272	6	270	6
severe	none	sm5	296	261	4	262	7
mixed	none	first	302	291	13	283	7
mixed	none	random	301	294	8	296	11
mixed	none	bn	302	297	10	297	10
mixed	none	bl	299	279	14	271	4
mixed	none	mean	303	303	12	303	13
mixed	none	sm.5	303	303	9	303	10
mixed	none	sm1	303	303	9	303	10
mixed	none	sm2	302	300	10	300	9
mixed	none	sm5	302	299	10	300	10
mixed	matched	first	302	291	13	284	7
mixed	matched	random	231	226	8	227	11
mixed	matched	bn	198	197	10	197	10
mixed	matched	bl	226	211	14	205	4
mixed	matched	mean	303	301	12	302	13
mixed	matched	sm.5	272	270	9	269	10
mixed	matched	sm1	224	223	9	223	10
mixed	matched	sm2	203	202	10	202	9
mixed	matched	sm5	198	198	10	198	10
mixed	mild	first	302	291	13	283	7
mixed	mild	random	231	226	8	227	11

Tier	Intruder	Policy	R1 /303	F known /303	F FA /143	A known /303	A FA /143
mixed	mild	bn	171	171	10	170	10
mixed	mild	bl	248	234	14	226	4
mixed	mild	mean	303	301	12	302	13
mixed	mild	sm.5	260	257	9	257	10
mixed	mild	sm1	205	204	9	204	10
mixed	mild	sm2	184	183	10	183	9
mixed	mild	sm5	175	174	10	174	10

A2. Complete temporal calibration operating points

Each threshold is selected from 66 unknown calibration tracks with zero accepted calibration queries. Values shown to 6 decimals are for reporting; original computation used full precision. Quality uses the fixed norm bounds 16-22; adaptive margin.08 and floor.20. These calibration budgets do not ensure low evaluation FPIR.

Tier	Intruder	Policy	Fixed threshold	Adaptive base
clean	none	first	0.210309	0.232873
clean	none	random	0.188370	-1.000000
clean	none	bn	0.196225	-1.000000
clean	none	bl	0.210309	0.232873
clean	none	mean	0.189518	-1.000000
clean	none	sm.5	0.190126	-1.000000
clean	none	sm1	0.190203	-1.000000
clean	none	sm2	0.194205	-1.000000
clean	none	sm5	0.196178	-1.000000
mild	none	first	0.228991	0.248854
mild	none	random	0.207864	0.238514
mild	none	bn	0.197199	-1.000000
mild	none	bl	0.228991	0.248854
mild	none	mean	0.203925	0.242105
mild	none	sm.5	0.202357	0.240537
mild	none	sm1	0.200642	0.238336
mild	none	sm2	0.201653	0.225331
mild	none	sm5	0.198258	-1.000000
medium	none	first	0.221121	0.235858
medium	none	random	0.225716	0.233878

Tier	Intruder	Policy	Fixed threshold	Adaptive base
medium	none	bn	0.231908	0.244665
medium	none	bl	0.221121	0.235858
medium	none	mean	0.231521	0.238094
medium	none	sm.5	0.231449	0.244206
medium	none	sm1	0.233905	0.246662
medium	none	sm2	0.233635	0.246392
medium	none	sm5	0.231400	0.244157
severe	none	first	0.274390	0.334839
severe	none	random	0.274390	0.334839
severe	none	bn	0.229365	0.271228
severe	none	bl	0.235794	0.285612
severe	none	mean	0.237205	0.281290
severe	none	sm.5	0.231067	0.270981
severe	none	sm1	0.227794	0.267708
severe	none	sm2	0.227273	0.267187
severe	none	sm5	0.230832	0.270746
mixed	none	first	0.209235	0.252628
mixed	none	random	0.225716	0.233146
mixed	none	bn	0.231908	0.246370
mixed	none	bl	0.211867	0.258457
mixed	none	mean	0.214824	0.229286
mixed	none	sm.5	0.222785	0.237247
mixed	none	sm1	0.225166	0.239628
mixed	none	sm2	0.225859	0.240322
mixed	none	sm5	0.227132	0.241594
mixed	matched	first	0.209235	0.252628
mixed	matched	random	0.225716	0.233146
mixed	matched	bn	0.231908	0.246370
mixed	matched	bl	0.211867	0.258457
mixed	matched	mean	0.214824	0.229286
mixed	matched	sm.5	0.222785	0.237247
mixed	matched	sm1	0.225166	0.239628
mixed	matched	sm2	0.225859	0.240322

Tier	Intruder	Policy	Fixed threshold	Adaptive base
mixed	matched	sm5	0.227132	0.241594
mixed	mild	first	0.209235	0.252628
mixed	mild	random	0.225716	0.233146
mixed	mild	bn	0.231908	0.246370
mixed	mild	bl	0.211867	0.258457
mixed	mild	mean	0.214824	0.229286
mixed	mild	sm.5	0.222785	0.237247
mixed	mild	sm1	0.225166	0.239628
mixed	mild	sm2	0.225859	0.240322
mixed	mild	sm5	0.227132	0.241594

A3. Complete paired temporal intervals

Change and limits are percentage points, computed from 1,000 identity-cluster draws. R1=Rank-1 effect; DIR=correct known acceptance under fixed rule. Help/harm counts refer to paired known tracks. Zero intervals can arise when all paired outcomes agree, not prove universal equivalence.

Tier/control	Comparison	Metric	Change	95% interval	Help/harm
clean/none	mean - first	R1	0.33	0.00 to 1.07	1/0
clean/none	mean - first	DIR	0.99	0.00 to 2.14	3/0
clean/none	bn - mean	R1	0.00	0.00 to 0.00	0/0
clean/none	bn - mean	DIR	-0.33	-0.97 to 0.00	0/1
clean/none	sm 1 - mean	R1	0.00	0.00 to 0.00	0/0
clean/none	sm 1 - mean	DIR	0.00	0.00 to 0.00	0/0
mild/none	mean - first	R1	0.00	0.00 to 0.00	0/0
mild/none	mean - first	DIR	1.65	0.33 to 3.43	5/0
mild/none	bn - mean	R1	0.00	0.00 to 0.00	0/0
mild/none	bn - mean	DIR	0.00	0.00 to 0.00	0/0
mild/none	sm 1 - mean	R1	0.00	0.00 to 0.00	0/0
mild/none	sm 1 - mean	DIR	0.00	0.00 to 0.00	0/0
medium/none	mean - first	R1	0.00	0.00 to 0.00	0/0
medium/none	mean - first	DIR	1.32	0.33 to 2.48	4/0
medium/none	bn - mean	R1	0.00	0.00 to 0.00	0/0
medium/none	bn - mean	DIR	-0.99	-2.57 to 0.00	0/3
medium/none	sm 1 - mean	R1	0.00	0.00 to 0.00	0/0

Tier/control	Comparison	Metric	Change	95% interval	Help/harm
medium/none	sm 1 - mean	DIR	-0.66	-1.60 to 0.00	0/2
severe/none	mean - first	R1	2.31	0.96 to 3.75	7/0
severe/none	mean - first	DIR	19.47	12.76 to 25.32	59/0
severe/none	bn - mean	R1	-2.64	-4.83 to -0.70	0/8
severe/none	bn - mean	DIR	-9.90	-13.74 to -5.69	0/30
severe/none	sm 1 - mean	R1	0.00	0.00 to 0.00	0/0
severe/none	sm 1 - mean	DIR	0.33	-1.38 to 1.97	3/2
mixed/none	mean - first	R1	0.33	0.00 to 0.99	1/0
mixed/none	mean - first	DIR	3.96	1.93 to 6.23	12/0
mixed/none	bn - mean	R1	-0.33	-0.99 to 0.00	0/1
mixed/none	bn - mean	DIR	-1.98	-5.08 to 0.00	0/6
mixed/none	sm 1 - mean	R1	0.00	0.00 to 0.00	0/0
mixed/none	sm 1 - mean	DIR	0.00	0.00 to 0.00	0/0
mixed/matched	mean - first	R1	0.33	0.00 to 0.99	1/0
mixed/matched	mean - first	DIR	3.30	1.66 to 5.24	10/0
mixed/matched	bn - mean	R1	-34.65	-42.91 to -26.07	0/105
mixed/matched	bn - mean	DIR	-34.32	-42.72 to -25.67	0/104
mixed/matched	sm 1 - mean	R1	-26.07	-33.87 to -18.48	0/79
mixed/matched	sm 1 - mean	DIR	-25.74	-34.15 to -17.98	0/78
mixed/mild	mean - first	R1	0.33	0.00 to 0.99	1/0
mixed/mild	mean - first	DIR	3.30	1.40 to 5.42	11/1
mixed/mild	bn - mean	R1	-43.56	-53.13 to -33.57	0/132
mixed/mild	bn - mean	DIR	-42.90	-52.52 to -33.00	0/130
mixed/mild	sm 1 - mean	R1	-32.34	-42.86 to -22.44	0/98
mixed/mild	sm 1 - mean	DIR	-32.01	-43.06 to -22.22	0/97

A4. Full reported runtime stage summaries

ENGINEERING. All 48 stage-summary rows are reproduced. Calls are repeated fixed inputs; stage medians are not additive. Microbenchmark values measure actual search and decision execution; other values are from image calls. The environment and timing boundaries are given in Chapters 3/4.

Model	Phase / stage	IDs	Calls	Median ms	p95 ms
adaface	enrollment / alignment	55	30	0.5963	1.0041
adaface	enrollment / decode	55	30	1.4223	16.2469

Model	Phase / stage	IDs	Calls	Median ms	p95 ms
adaface	enrollment / detection	55	30	89.2764	95.4250
adaface	enrollment / embedding	55	30	104.0582	113.4520
adaface	enrollment / enrollment_end_to_end	55	30	199.8750	221.3193
adaface	enrollment / padding	55	30	0.6924	9.9648
adaface	enrollment / persistent_insertion	55	30	1.5802	2.1713
adaface	enrollment / quality	55	30	0.2147	0.2828
adaface	enrollment / template_normalization	55	30	0.0136	0.0159
adaface	microbenchmark / decision	10	300	0.0012	0.0015
adaface	microbenchmark / decision	25	300	0.0012	0.0013
adaface	microbenchmark / decision	55	300	0.0012	0.0013
adaface	microbenchmark / gallery_search	10	300	0.0293	0.0402
adaface	microbenchmark / gallery_search	25	300	0.0408	0.0438
adaface	microbenchmark / gallery_search	55	300	0.0635	0.0695
adaface	recognition / alignment	55	60	0.7272	1.2193
adaface	recognition / decision	55	60	0.0038	0.0043
adaface	recognition / decode	55	60	8.3162	25.7682
adaface	recognition / detection	55	60	88.2088	97.9901
adaface	recognition / embedding	55	60	103.7119	113.0020
adaface	recognition / gallery_search	55	60	0.1299	0.1541
adaface	recognition / padding	55	60	3.0315	11.1618
adaface	recognition / quality	55	60	0.2287	0.3189
adaface	recognition / recognition_end_to_end	55	60	207.0253	241.8298
arcface	enrollment / alignment	55	30	0.6289	1.0315
arcface	enrollment / decode	55	30	1.1346	12.6483
arcface	enrollment / detection	55	30	86.4162	91.9729
arcface	enrollment / embedding	55	30	44.0791	58.3522
arcface	enrollment / enrollment_end_to_end	55	30	136.9683	173.1506
arcface	enrollment / padding	55	30	0.6289	8.1421
arcface	enrollment / persistent_insertion	55	30	1.8721	2.7116
arcface	enrollment / quality	55	30	0.2402	0.2912
arcface	enrollment / template_normalization	55	30	0.0099	0.0167
arcface	microbenchmark / decision	10	300	0.0011	0.0015
arcface	microbenchmark / decision	25	300	0.0011	0.0013

Model	Phase / stage	IDs	Calls	Median ms	p95 ms
arcface	microbenchmark / decision	55	300	0.0012	0.0014
arcface	microbenchmark / gallery_search	10	300	0.0296	0.0438
arcface	microbenchmark / gallery_search	25	300	0.0408	0.0421
arcface	microbenchmark / gallery_search	55	300	0.0637	0.0708
arcface	recognition / alignment	55	60	0.7569	1.1863
arcface	recognition / decision	55	60	0.0044	0.0052
arcface	recognition / decode	55	60	7.2172	29.1962
arcface	recognition / detection	55	60	86.3946	92.5805
arcface	recognition / embedding	55	60	44.1930	55.0558
arcface	recognition / gallery_search	55	60	0.1542	0.1711
arcface	recognition / padding	55	60	2.5639	10.8869
arcface	recognition / quality	55	60	0.2465	0.3042
arcface	recognition / recognition_end_to_end	55	60	143.8087	175.5687

A5. Historical sequential enrollment demonstration

This older dynamic-gallery demonstration differs from fixed-probe growth: the enrolled-known denominator changes as IDs are inserted. Use it as functionality evidence. Do not compare its changing denominators as an accuracy-versus-gallery-size experiment.

Gallery IDs	Active known attempts	Correct known	Not enrolled attempts	False accepts	Search median ms
0	0	0	304	0	0.0010
10	24	23	280	0	0.0290
20	50	47	254	0	0.0360
30	80	72	224	1	0.0435
40	110	100	194	1	0.0515
55	154	139	150	1	0.0631

A6. Development operating thresholds

The next tables remove any need to look up selected operating points in external files. Backbone values are full-precision saved settings. Enrollment-variant values are shown to 9 decimal places for reporting. These are development-calibrated choices, not thresholds selected on the final identities. Growth uses each checkpoint's up-to-three-photo backbone settings unchanged at every composition.

Checkpoint	Enrollment	Rule	Selected score threshold	Gap
arcface	One original	Score	0.3913350999355316	0.0

Checkpoint	Enrollment	Rule	Selected score threshold	Gap
arcface	One original	Gap	0.3206184208393097	0.1
arcface	Up to 3 originals	Score	0.3978477716445923	0.0
arcface	Up to 3 originals	Gap	0.3778589367866516	0.15
adaface	One original	Score	0.3992882966995239	0.0
adaface	One original	Gap	0.32279378175735474	0.1
adaface	Up to 3 originals	Score	0.4159545302391052	0.0
adaface	Up to 3 originals	Gap	0.33722615242004395	0.15

Checkpoint	Enrollment	View / rule	Score threshold	Gap
arcface	One original	Fixed / Score	0.391335100	0.0
arcface	One original	Recal / Score	0.391335100	0.0
arcface	One original	Fixed / Gap	0.320618421	0.1
arcface	One original	Recal / Gap	0.320618421	0.1
arcface	One+5 variants	Fixed / Score	0.391335100	0.0
arcface	One+5 variants	Recal / Score	0.396722227	0.0
arcface	One+5 variants	Fixed / Gap	0.320618421	0.1
arcface	One+5 variants	Recal / Gap	0.331417024	0.15
arcface	Up to 3 originals	Fixed / Score	0.391335100	0.0
arcface	Up to 3 originals	Recal / Score	0.397847772	0.0
arcface	Up to 3 originals	Fixed / Gap	0.320618421	0.1
arcface	Up to 3 originals	Recal / Gap	0.377858937	0.15
adaface	One original	Fixed / Score	0.399288297	0.0
adaface	One original	Recal / Score	0.399288297	0.0
adaface	One original	Fixed / Gap	0.322793782	0.1
adaface	One original	Recal / Gap	0.322793782	0.1
adaface	One+5 variants	Fixed / Score	0.399288297	0.0
adaface	One+5 variants	Recal / Score	0.430022418	0.0
adaface	One+5 variants	Fixed / Gap	0.322793782	0.1
adaface	One+5 variants	Recal / Gap	0.323049664	0.1
adaface	Up to 3 originals	Fixed / Score	0.399288297	0.0
adaface	Up to 3 originals	Recal / Score	0.415954530	0.0
adaface	Up to 3 originals	Fixed / Gap	0.322793782	0.1
adaface	Up to 3 originals	Recal / Gap	0.337226152	0.15

All above quality gates are 0. The development original-photo pooling fixed-gap sensitivity uses tau=.3778589367866516 and gap=.15 for every policy; its recalibrated score thresholds are separately selected using the declared unknown-score budget. Robustness uses the original decision-family thresholds without retuning. Do not mix a fixed-rule comparison with separately calibrated backbone choices.

A7. Complete temporal pairwise verification summaries

EXPLORATORY descriptive pairwise results, not independently calibrated identification thresholds. For every row, 303 known and 143 unknown tracks are scored against 17 gallery templates: 303 genuine pairs and 7,279 impostor pairs. FAR1% and FAR0.1% are pair denominators; the main FA tables instead use 143 unknown queries. The condition/control/policy abbreviations are the same as A1.

Tier/control	Policy	TAR@1%	TAR@0.1%	EER	ROC AUC
clean/none	first	99.01%	89.77%	0.99%	0.99939
clean/none	random	99.01%	88.12%	0.99%	0.99950
clean/none	bl	99.01%	87.13%	0.99%	0.99940
clean/none	bn	99.34%	90.43%	0.98%	0.99955
clean/none	mean	99.67%	89.44%	0.71%	0.99961
clean/none	sm.5	100.00%	90.43%	0.92%	0.99962
clean/none	sm1	99.67%	90.10%	0.67%	0.99960
clean/none	sm2	99.67%	89.44%	0.93%	0.99956
clean/none	sm5	99.34%	90.10%	0.93%	0.99956
mild/none	first	99.01%	86.80%	0.99%	0.99939
mild/none	random	99.67%	86.80%	0.93%	0.99948
mild/none	bl	99.67%	87.13%	0.73%	0.99950
mild/none	bn	99.01%	89.77%	0.99%	0.99952
mild/none	mean	100.00%	88.12%	0.91%	0.99957
mild/none	sm.5	99.67%	89.11%	0.92%	0.99959
mild/none	sm1	99.67%	89.11%	0.91%	0.99958
mild/none	sm2	99.34%	89.44%	0.73%	0.99955
mild/none	sm5	99.34%	89.44%	0.99%	0.99952
medium/none	first	98.35%	81.19%	1.04%	0.99916
medium/none	random	99.34%	81.85%	0.96%	0.99931
medium/none	bl	99.01%	82.51%	0.99%	0.99930
medium/none	bn	99.67%	81.85%	0.74%	0.99941
medium/none	mean	99.67%	80.53%	0.72%	0.99947
medium/none	sm.5	100.00%	83.83%	0.73%	0.99948
medium/none	sm1	99.67%	84.16%	0.94%	0.99949

Tier/control	Policy	TAR@1%	TAR@0.1%	EER	ROC AUC
medium/none	sm2	99.67%	84.16%	0.94%	0.99946
medium/none	sm5	99.67%	83.83%	0.96%	0.99944
severe/none	first	89.11%	66.67%	4.93%	0.99303
severe/none	random	89.44%	59.08%	5.30%	0.98973
severe/none	bl	88.78%	62.71%	4.29%	0.99201
severe/none	bn	88.78%	63.37%	4.29%	0.99404
severe/none	mean	95.38%	65.02%	2.37%	0.99735
severe/none	sm.5	94.72%	64.36%	2.31%	0.99715
severe/none	sm1	94.72%	63.04%	2.93%	0.99667
severe/none	sm2	92.41%	62.38%	3.62%	0.99551
severe/none	sm5	90.10%	61.72%	3.96%	0.99445
mixed/none	first	95.38%	81.52%	2.27%	0.99811
mixed/none	random	97.36%	79.54%	1.97%	0.99821
mixed/none	bl	92.08%	69.64%	2.64%	0.99614
mixed/none	bn	98.35%	89.11%	1.03%	0.99907
mixed/none	mean	100.00%	89.44%	0.66%	0.99963
mixed/none	sm.5	100.00%	88.78%	0.66%	0.99962
mixed/none	sm1	100.00%	89.11%	0.92%	0.99957
mixed/none	sm2	99.01%	87.79%	0.99%	0.99932
mixed/none	sm5	99.01%	86.47%	0.99%	0.99915
mixed/matched	first	95.38%	81.52%	2.27%	0.99811
mixed/matched	random	62.71%	5.28%	18.81%	0.87606
mixed/matched	bl	59.41%	2.31%	22.12%	0.85843
mixed/matched	bn	40.59%	3.30%	25.08%	0.82475
mixed/matched	mean	97.69%	85.15%	1.32%	0.99905
mixed/matched	sm.5	89.44%	48.18%	2.60%	0.99605
mixed/matched	sm1	68.32%	14.19%	5.66%	0.97877
mixed/matched	sm2	49.50%	6.60%	13.22%	0.92166
mixed/matched	sm5	44.55%	5.61%	20.79%	0.86485
mixed/mild	first	95.38%	81.52%	2.27%	0.99811
mixed/mild	random	62.05%	4.95%	19.47%	0.86964
mixed/mild	bl	74.59%	2.97%	15.18%	0.90567
mixed/mild	bn	25.74%	1.65%	30.03%	0.77094

Tier/control	Policy	TAR@1%	TAR@0.1%	EER	ROC AUC
mixed/mild	mean	97.69%	84.16%	1.63%	0.99902
mixed/mild	sm.5	87.13%	42.90%	3.01%	0.99503
mixed/mild	sm1	54.46%	8.91%	7.59%	0.97410
mixed/mild	sm2	38.94%	2.64%	18.15%	0.88784
mixed/mild	sm5	30.36%	1.98%	25.72%	0.81546

References, citation placement and source record

The entries below support the material actually used in this guide. Cite them in the thesis where the relevant method, dataset, application or mapping is introduced. Source summaries are paraphrased. Do not copy the old report's unresolved references or template example as evidence. The listed web resources were checked during preparation; repository references describe software documentation, not reviewed comparative recognition results.

[R1] J. Deng,J. Guo,N. Xue,and S. Zafeiriou. ArcFace: Additive Angular Margin Loss for Deep Face Recognition. CVPR,2019;author preprint version3.

https://arxiv.org/abs/1801.07698v3

Use: Cite for embedding background and the pretrained ArcFace method,not as your own training result.

[R2] M. Kim,A. K. Jain,and X. Liu. AdaFace: Quality Adaptive Margin for Face Recognition. CVPR,2022.

https://arxiv.org/abs/2204.00964

Use: Cite for AdaFace training concept and released comparative checkpoint; distinguish shared SCRFD preprocessing.

[R3] J. Guo,J. Deng,A. Lattas,and S. Zafeiriou. Sample and Computation Redistribution for Efficient Face Detection. arXiv:2105.04714,2021.

https://arxiv.org/abs/2105.04714

Use: Cite SCRFD;do not borrow published detector AP/speed as your measurement.

[R4] R. Ranjan,A. Bansal,H. Xu,S. Sankaranarayanan,J.-C. Chen,C. D. Castillo,and R. Chellappa. Crystal Loss and Quality Pooling for Unconstrained Face Verification and Recognition. arXiv:1804.01159,2018,updated2019.

https://arxiv.org/abs/1804.01159

Use: Cite prior quality-pooling work;no first-in-literature pooling claim.

[R5] A. Bewley,Z. Ge,L. Ott,F. Ramos,and B. Upcroft. Simple Online and Realtime Tracking. ICIP,2016. DOI:10.1109/ICIP.2016.7533003.

https://arxiv.org/abs/1602.00763

Use: Cite SORT background;the local tracker is SORT-like and has no tracking benchmark.

[R6] T. Ghosh,Md. J. Al Nahian,Md. H. Al Banna,M. S. Kaiser,S. Al Mamun,and M. Mahmud. halfFace: A face covering mask dataset of South Asian people. Mendeley Data,version2,2022. DOI:10.17632/pk44mkx9vm.2.

https://data.mendeley.com/datasets/pk44mkx9vm/2

Use: Cite source dataset and documented labels;local926-image audit is this project's result.

[R7] Y. Wong,S. Chen,S. Mau,C. Sanderson,and B. C. Lovell. Patch-based Probabilistic Image Quality Assessment for Face Selection and Improved Video-based Face Recognition. IEEE Biometrics Workshop,CVPR Workshops,pp.81-88,2011.

https://arma.sourceforge.net/chokepoint/

Use: Attribute ChokePoint/NICTA and identify the custom supplied-crop protocol. Derivatives are labeled synthetic.

[R8] A. Bhattacharjee and S. Das. D2SC-GAN: Dual Deep-Shallow Channeled Generative Adversarial Network,for Resolving Low-Resolution Faces for Recognition in Classroom Scenarios. IEEE TBIOM2(3),223-234,2020. DOI:10.1109/TBIOM.2020.2983524.

https://sites.google.com/view/icfd

Use: Cite ICFD context. Only the public sample was evaluated;no GAN was trained.

[R9] Exadel. CompreFace: open-source face-recognition service. Project documentation,accessed23 October2026.

https://github.com/exadel-inc/CompreFace

Use: Cite similar application functionality;no head-to-head benchmark.

[R10] S. I. Serengil and contributors. DeepFace project documentation,accessed23 October2026.

https://github.com/serengil/deepface

Use: Cite software/application context;not an independent product-superiority result.

[R11] Board of Accreditation for Engineering and Technical Education. ACC-MAN-02 Accreditation Criteria,version2.2,effective1 July2024. Tables5.8.1-5.8.3.

https://www.baetebangladesh.org/acc-man-02-v2-2-f.html

Use: Legacy K/P/A labels correspond to supplied template;the project mappings are reasoned proposals.

[R12] Board of Accreditation for Engineering and Technical Education. ACC-MAN-02 Accreditation Criteria,version3.0,effective1 July2025.

https://baetebangladesh.org/acc-man-02-v3-f.html

Use: Documents newer WK/WP/EA terminology;do not silently replace the department template or claim accreditation.

[R13] Deep Insight contributors. InsightFace model zoo documentation,accessed23 October2026.

https://github.com/deepinsight/insightface/tree/master/model_zoo

Use: Cite released model bundle/provenance;model terms differ from source-code licensing.

[R14] M. Kim and contributors. Official AdaFace implementation;source commit c60eaa786a42c03444f3df7096dbaf9d57ae010d.

https://github.com/mk-minchul/AdaFace

Use: Cite source/checkpoint integration and official preprocessing.

[T1] United International University,Department of CSE. FYDP report template supplied by the user,17-page PDF,example title date22 October2026.

Use: The format authority for this guide;not an instruction to fabricate missing project facts.

Retain a consistent institutional bibliography style and number citations automatically. ArcFace 2019 version 3 has four authors; do not mix it with the later six-author journal-version metadata. Publication dates are not local experiment dates. No fabricated DOI, venue, page range or publication status should be added. The existing narrow core review is not represented as a systematic review of every surveillance paper.

Source rights and data handling

ChokePoint is attributed to NICTA and its required research reference; its source terms restrict commercial use and require derivative labeling. InsightFace model-zoo assets are documented for non-commercial research; do not assume the library code license grants all model uses. AdaFace source is MIT-licensed, but checkpoint/pretraining permissions remain distinct. Public research access does not create institutional permission to collect new student data or commercialize a monitoring service. The guide contains aggregate charts and photo-hidden interface screenshots; no raw face crops or embedding arrays are distributed.

Evidence provenance and checksums

The following identifiers are for audit traceability, not instructions to leave this PDF to obtain the writing content. Methods, result fractions, scope and relevant reporting-level tables are embedded above.

Main final: results/final/heldout_20261019T231512869782Z

Backbone development: results/backbones/backbones_development_20261020T200004286474Z

Enrollment extensions: results/enrollment_extensions/extensions_20261022T193101706934Z

ChokePoint supplementary: results/chokepoint/temporal_20261022T185858644292Z

Pooling development: results/enrollment_pooling/pooling_development_20261020T181729489649Z

Decision development: results/decisions/decisions_development_20261019T222927022126Z

Robustness development: results/robustness/robustness_development_20261019T230159679928Z

Asset	SHA-256
ArcFace w 600 k_r 50.onnx	4c06341c33c2ca1f86781dab0e829f88ad5b64be9fba56e56bc9ebdefc619e43
SCRFD det_10 g.onnx	5838f7fe053675b1c7a08b633df49e7af5495cee0493c7dcf6697200b85b5b91
AdaFace IR-50/MS1 MV2	234da6ce931f821e0a8e920063dc2091d8122ea46285c159d2f25d3f762fecb3
Main freeze receipt	24b8a269fe27a6327d4be08e315df61996318ab713e6e55edfbcbb23158fab94
Original UIU template	3be2fa41560a2104ad30878b460ca22739d0ed5c5b6fab0634402acaa860fca2

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Writing integration, figure use and final completeness check

What to remove from the previous report

- FYDP1-only future-tense abstract and statements that implementation is deferred, when describing the present final study.
- Haar/Sobel/Canny/40 x 40 pipeline and seven-class CNN50-epoch training diagrams as the current method. If preserved as historical work, label them historical and resolve old numerical conflicts.
- PostgreSQL users/logs architecture as the current storage design; current persistence is JSON/NPZ.
- The old 2 x 2 confusion matrix and inconsistent accuracy/precision claims. Old PCA50 values 51.55% versus 54.65% disagree; CNN46.12% final test is not 91.09% peak validation.
- Table 4.7's TP226, FN32, TN400, FP0 would imply 95.14% binary accuracy and 1.00 precision, not 96.23% and.36. Its figure places 400 in unknown-to-known despite a zero-error narrative. Do not repair by guessing without old raw predictions.
- Claims of a new trained adapter, quality network, universally superior weighting, validated real-time camera capacity, demographic fairness, anti-spoofing or institution-wide security improvement.

Embedded figure and table placement

Material in this PDF	Proposed thesis location	How to use it
Figures 3.1-3.3 context, DFD, UI	Sections3.1.2-3.1.4	Copy/redraw the embedded vector figure; keep its implemented/prototype labels.
Figures 3.4-3.7 audit, pipeline, rule, freeze	Detailed 3.2 methods	Preserve branches, roles and capture-failure distinction.
Figures 3.8-3.9 temporal and lifecycle	Detailed 3.2 / supplemental design	Do not merge crop aggregation with a tested live tracker.
Figures 4.1-4.5 final, stress, variants, growth, temporal	Corresponding 4.3 result subsections	Retain scope and denominator captions; do not relabel development as final.
Figures 4.6-4.7 historical UI screenshots	UI implementation/demo subsection	Disclosed development examples; not fresh camera capture.
Figure 4.8 CPU timing	Runtime subsection	State excluded stages and engine differences.
Tables4.1-4.21	Main results / selected appendices	Use fractions plus rates; do not remove refusals/failures.
Tables 5.1/5.2 and P1 knowledge table	Sections5.4.1-5.4.2	Keep qualification and actual-project rationale; not blanket ticks.
Supplementary A1-A7	Appendix if long tables exceed body space	Complete reporting-level outputs and operating points remain here; no external spreadsheet required.

The PDF's diagrams are embedded vector artwork; copy a figure from a PDF editor or redraw it from its visible labels/arrows. A screenshot/crop is possible for a draft, but use a clear resolution and preserve the caption. Do not make a portrait/contact sheet just to decorate the report. Final thesis figure numbering must be regenerated if the order changes.

Author-supplied facts that cannot be manufactured

Input needed	Action in final thesis
Correct names/IDs and supervisor designation	Confirm official student records and supervisor details; do not treat transcription as verification.
Submission date and required institutional forms	Use actual date and genuine approved/signature material. The supplied PDF does not specify every possible separate administrative form.
Actual acknowledgements/publications	Thank real contributors; omit optional publication list if none.
Historical member contributions and work dates	Fill task allocation from actual records; the proposed writing roles are not authorship evidence.
Actual expenditure and deployment quotations	Replace hypothetical budget amounts with supported amounts or explicitly retain them as planning examples.
Institutional choice of mapping labels/word limits	The guide follows the supplied P/A template; any separate later departmental instruction takes precedence.
Supervisor review / final title approval	Review the proposed title, scope and final manuscript; do not imply existing approval.

No further technical file is needed to follow the writing guide. These personal/administrative facts are the unavoidable remaining author inputs; inventing them would make the report less complete and less credible. A university's separately issued requirements cannot be guaranteed from a 17-page supplied template alone.

Template coverage matrix

Template item	Coverage in this guide
Title/Abstract/Acknowledgements	All included, with single-paragraph abstract and author-input limits.
Publication List	Explicitly optional; truthful omission/no-publication treatment included.
Contents/Figures/Tables lists	Generation, numbering and caption instructions included.
1.1-1.6	Overview,motivation,objectives,methodology,outcome,organization.
2.1	Preliminaries and terminology.
2.2/2.2.1/2.2.2	Literature, similar applications and related research.
2.3/2.4	Gap analysis and chapter summary.
3.1/3.1.1	Requirements, including functional and nonfunctional.
3.1.2	Context diagram embedded.
3.1.3	Level 1 data-flow diagram embedded.
3.1.4	UI design and wireframe embedded.
3.2	Detailed methods, equations, alternatives and choice rationale.
3.3/3.4/3.5	Project plan, task allocation and summary.
4.1	Environment with actual execution boundaries.
4.2	Software tests, numeric audits and evaluation setup.
4.3	Final, development, supplementary and engineering results/discussion.
4.4	Implementation/results summary.
5.1.1-5.1.3	Software, hardware and communication standards/alternatives.
5.2.1-5.2.7	Economic,environmental,ethical,health/safety,social,political,sustainability.
5.3	Budget, alternative budget, rationales and revenue-model treatment.
5.4.1	P1-P7 table, each rationale, and additional P1 knowledge-profile mapping.
5.4.2	A1-A5 table and individual rationale guidance.
5.5	Standards/constraints summary.
6.1/6.2/6.3	Summary, limitation and future work.
References	Core complete citation entries and placement guidance supplied.
Chapter outlines	One/two-sentence draft at the start of each of the six chapters.

Final editorial and numerical checklist

- Use the same title and scope throughout; no abstract claim exceeds its result section.
- Retain every original template top-level heading; use one/two outline sentences at every chapter start.

- Abstract is a single paragraph; all author-input fields are resolved or inapplicable optional material removed.
- Every figure/table is embedded, numbered, captioned and discussed in the text; lists/contents/page numbering are refreshed.
- Methods define template construction, preprocessing, calibration, ties, probe roles, failure accounting and uncertainty before results.
- Use FINAL/DEVELOPMENT/EXPLORATORY/ENGINEERING labels consistently; never merge the 154-known development denominator with 160-known final.
- Report 151/160=94.38%, not 159/160 retrieval as the primary acceptance figure. Report 0/151 as observed, not a population guarantee.
- Margin benefit is one net probe with interval including zero. No claim of quality-gate improvement when gate=0.
- Unknown FA denominators are valid unknown probes where conditional FPIR is shown; capture failures and refusals remain visible.
- Synthetic variants/frames are not additional independent people. Growth composition ranges are not confidence intervals.
- Adaptive threshold diagnostic is separate from main decisions; pairwise FAR is not identification FPIR.
- Timing values are direct measurements with excluded stages; no camera FPS, energy saving or arbitrary-scale claim is invented.
- Alternative design/standards rationales, communication/political constraints, budget alternative, revenue treatment, P1-K mapping andA1-A5 rationales are present.
- Old inaccurate tables, confusion matrix, placeholder captions, unresolved citations and obsolete PostgreSQL/CNN-current-method claims are removed or corrected with evidence.
- References have consistent metadata and in-text citations; no publication, approval, team role, cost or demographic claim is fabricated.
- All authors review the conclusion, limitations and responsibilities; supervisor and any separate administrative requirements are handled honestly.

Concise defense answers for the writing team

Question	Evidence-backed answer
What did you contribute?	The declared paired experimental investigation, failure-aware accounting, analysis of limits and working persistent gallery; not a new backbone.
Why not claim 100% or 99.38%?	99.38% is top-match retrieval; primary correct acceptance is 94.38% with refusals/failures included.
Did the margin clearly improve performance?	Only one net probe beyond few-photo score; paired interval includes zero.
Does zero false acceptance make it safe?	No; 0/151 observed probes from 27 unknown IDs cannot establish a population guarantee.
Is AdaFace better?	The released development checkpoints show different acceptance/coverage/errors and engines; no isolated universal winner.
Why include negative augmentation/weighting results?	They answer the declared operating-choice questions and avoid selecting only favorable outcomes.
Was CCTV evaluated end to end?	No; still-image pipeline and supplied-crop temporal analysis are separate. Live camera/tracker operation needs its own benchmark.
What about ViT and adapter?	FaceLiVT integration remains unfinished; adapter training is toy synthetic-vector work, not validated real-domain adaptation.
What is needed next?	Independent representative data, calibration transfer, live-system measurements and institutionally appropriate governance/security.

Completion statement. This guide covers every heading and explicit instruction visible in the supplied original UIU template and embeds the research information needed to draft those sections. It does not certify institutional approval or invent the author-specific facts listed above. Preserve that distinction in the final thesis.

Appendix B. Original supplied UIU template

The next 17 pages reproduce the original UIU Department of CSE FYDP template inside this same PDF. They preserve its title-page arrangement, logo, preliminary pages, chapter headings and instruction tables so no separate template file is needed for reference.

Each reproduced page has a small REFERENCE TEMPLATE header identifying its original page and combined-PDF page. Original printed Roman/Arabic page numbers are retained as part of the supplied reference. Use the PDF bookmarks to jump to individual template pages.

How to apply it. Use the original arrangement and heading hierarchy when preparing the final thesis. Replace example title, student name/date and placeholder instructions with confirmed project content. The guide chapters above supply the actual methods, results, diagrams and writing decisions for those slots. In particular, the old report's subsection numbering must not override this original template.

Reference status. The reproduced template's example Kleinberg/Tardos reference, blank publication entries and sample check marks are illustrations, not new evidence about your project. The template's requirement to explain alternatives, budget/revenue assumptions and engineering mappings is addressed explicitly in the guide. No signature, approval or publication has been manufactured.

> The presence of the original pages is for complete in-file guidance. Do not append blank instruction pages to your final submitted thesis unless the university explicitly asks for them.

Title of Your
Final Year Design Project

By

Student Name and ID

Submitted in partial fulfilment of the requirements
of the degree of Bachelor of Science in Computer Science and Engineering

October 22, 2026

Department of Computer Science and Engineering
United International University

Abstract

The abstract should contain a summary of the work presented in this report in a single paragraph. The paragraph should provide a general background of the problem, methodology and results.

Acknowledgements

This work would have not been possible without the input and support of many people over the last two trimesters. We would like to express my gratitude to everyone who contributed to it in some way or other.

First, we would like to thank my academic advisors, ...

Our sincere gratitude goes to ...

We are also thankful to ...

Last but not the least, We owe to our family including our parents for their unconditional love and immense emotional support.

Publication List

[Optional] The main contributions of this research are either published or accepted or in preparation in journals and conferences as mentioned in the following list:

Journal Articles

1.

Conference Papers

1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

1.

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Chapter 1

Introduction

[*Must be present in FYDP-1 Report and also in Final Report*]

Every chapter should start with 1-2 sentences on the outline of the chapter.

1.1 Project Overview

This section should clearly present the background and a problem statement that your project aims to solve.

1.2 Motivation

The computational motivation that encourages you to solve the problem should be stated here clearly. In addition, you can mention why solving this problem will benefit.

1.3 Objectives

Enumerate the objectives in clear and specific terms.

1.4 Methodology

Here you put a brief summary of the methodology.

1.5 Project Outcome

What are or could be the possible outcomes of your work?

1.6 Organization of the Report

Here put a chapter wise structure of the report in narrative form.

Chapter 2

Background

[*Must be present in FYDP-1 Report and also in Final Report*]

Every chapter should start with 1-2 sentences on the outline of the chapter.

2.1 Preliminaries

In this section, you have to provide the necessary background knowledge to understand the rest of the report [1].

2.2 Literature Review

This section will contain you literature review.

2.2.1 Similar Applications

Put a summary of similar web applications, mobile apps similar to your work.

2.2.2 Related Research

Here report the summary of the investigation of the research literature.

2.3 Gap Analysis

Here summarise the gap where you intend to work.

2.4 Summary

Chapter 3

Project Design

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

3.1 Requirement Analysis

3.1.1 Functional and Nonfunctional Requirements

3.1.2 Context Diagram

3.1.3 Data Flow Diagram Level 1

3.1.4 UI Design

3.2 Detailed Methodology and Design

You have to mention alternate solutions that you have considered. Why you have selected the specific solution, etc.

3.3 Project Plan

3.4 Task Allocation

3.5 Summary

Chapter 4

Implementation and Results

[Must be present in Final Report. Incomplete version might be included in FYDP-1 Report, however it is optional.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

4.1 Environment Setup

4.2 Testing and Evaluation

4.3 Results and Discussion

4.4 Summary

Chapter 5

Standards and Design Constraints

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

5.1 Compliance with the Standards

Only mention the standards that are related to your project. This list is not complete. For each of the standards discuss the alternates with pros and cons and rationale of selection.

5.1.1 Software Standards

5.1.2 Hardware Standards

5.1.3 Communication Standards

5.2 Design Constraints

Only mention the constraints that are related to the design of your project. This list is not complete.

5.2.1 Economic Constraint

5.2.2 Environmental Constraint

5.2.3 Ethical Constraint

5.2.4 Health and Safety Constraint

5.2.5 Social Constraint

5.2.6 Political Constraint

5.2.7 Sustainability

5.3 Cost Analysis

Provide a cost analysis in terms of budget required and revenue model. In case of budget, you must show an alternate budget and rationales.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 5.1). For P1, you need to put another mapping with Knowledge profile and rational thereof.

Table 5.1: Mapping with complex problem solving.

P1 Dept of Knowledge	P2 Range of Conflicting Requirements	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stakeholder Involvement	P7 Inter-dependence
√	√					

5.4.2 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 5.2).

5.5 Summary

Table 5.2: Mapping with complex engineering activities.

A1 Range of resources	A2 Level of Interaction	A3 Innovation	A4 Consequences for society and environment	A5 Familiarity
√	√			

Chapter 6

Conclusion

[Must be present in FYDP-1 Report and also in Final Report. Might be incomplete in FYDP-1 Report.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

6.1 Summary

6.2 Limitation

6.3 Future Work

References

[1] Jon Kleinberg and Eva Tardos. *Algorithm design*. Pearson Education India, 2006.